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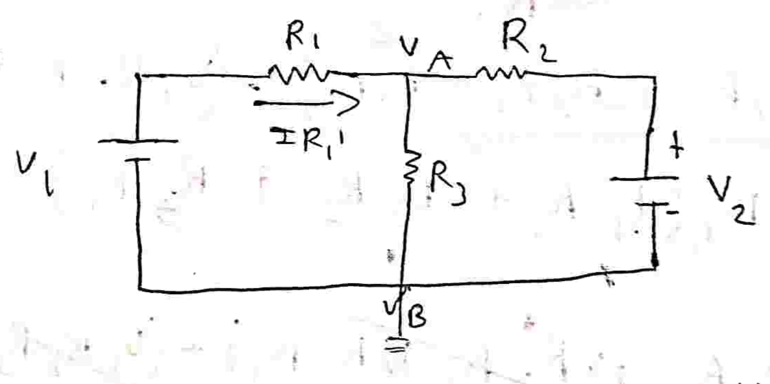
superposition Theorem :

statement :

In any linear Bilateral network containing more than one independent source (V or I) The response (current through or voltage across) In any element is the algebraic sum of all the responses produced by the sources acting alone

Note : To find individual responses voltage sources are shorted and current sources are open ~~ckts~~ circuited :

- consider the following network



Let the response is current through R1 (IR1)

$$\frac{V_A - V_1}{R_1} + \frac{V_A}{R_3} + \frac{V_A - V_2}{R_2} = 0$$

$$V_A\left(\frac{1}{R_1} + \frac{1}{R_3} + \frac{1}{R_2}\right) = \frac{V_1}{R_1} + \frac{V_2}{R_2}$$

$$V_A\left(\frac{1}{R_1} + \frac{1}{R_3} + \frac{1}{R_2}\right) = \frac{V_1 R_2 + V_2 R_1}{R_1 + R_2}$$

$$V_A\left(\frac{R_3R_2+R_1R_2+R_1R_3}{R_1R_2R_3}\right)=\left(\frac{V_1R_2+V_2R_1}{R_1R_2}\right)$$

$$V_A=\left(\frac{\cancel{R_1}\cancel{R_2}R_3}{R_3R_2+R_1R_2+R_1R_3}\right)\left(\frac{V_1R_2+V_2R_1}{\cancel{R_1}\cancel{R_2}}\right)$$

$$\boxed{V_A=\frac{V_1R_2R_3+V_2R_1R_3}{R_1R_2+R_3R_2+R_1R_3}} \quad \ldots (1)$$

The response is $I_{R_1}=\dfrac{V_1-V_A}{R_1}$ ... (2)

subs (1) in (2)

$$I_{R_1}=\frac{V_1-\dfrac{V_1R_2R_3+V_2R_1R_3}{R_1R_2+R_3R_2+R_1R_3}}{R_1}$$

$$I_{R_1}=\frac{V_1(R_1R_2+R_3R_2+R_1R_3)-(V_1R_2R_3+V_2R_1R_3)}{R_1(R_1R_2+R_2R_3+R_3R_1)}$$

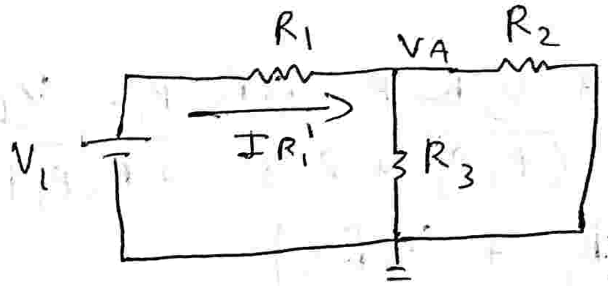
$$I_{R_1}=\frac{(V_1R_1R_2+\cancel{V_1R_3R_2}+V_1R_1R_3)-(\cancel{V_1R_2R_3}+V_2R_1R_3)}{R_1(R_1R_2+R_2R_3+(R_1R_3)}$$

$$I_{R_1}=\frac{\cancel{R_1}(V_1R_2+V_1R_3-V_2R_3)}{\cancel{R_1}(R_1R_2+R_2R_3+R_3R_1)}$$

$$\boxed{I_{R_1}=\frac{V_1(R_2+R_3-V_2R_3)}{R_1R_2+R_2R_3+R_3R_1}} \quad \ldots (A)$$

To prove $I R_1 = I R_1' + I R_1''$

case 1: V_1 acting alone



apply KCL

$$\frac{V_A - V_1}{R_1} + \frac{V_A}{R_3} + \frac{V_A}{R_2} = 0$$

$$V_A\left(\frac{1}{R_1} + \frac{1}{R_3} + \frac{1}{R_2}\right) = \frac{V_1}{R_1}$$

$$V_A\left(\frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1 R_2 R_3}\right) = \frac{V_1}{R_1}$$

$$V_A = \frac{V_1 \cancel{R_1} R_2 R_3}{\cancel{R_1}(R_1R_2 + R_2 \cdot R_3 + R_3R_1)}$$

$$\therefore V_A = \frac{V_1 R_2 R_3}{(R_1R_2 + R_2R_3 + R_3R_1)} \quad \text{--- } (3)$$

from the ckt

$$I R_1' = \frac{V_1 - V_A}{R_1}$$

$$I R_1' = \frac{V_1 - \dfrac{V_1 R_2 R_3}{R_1R_2 + R_2R_3 + R_3R_1}}{R_1}$$

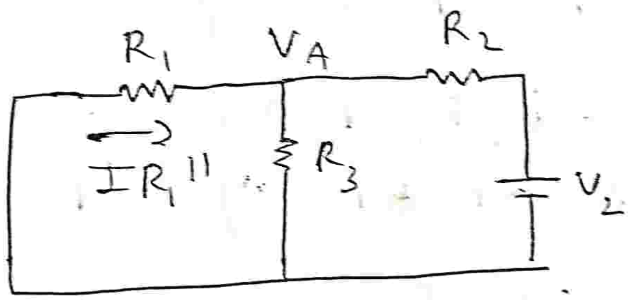
$$IR_1' = \frac{V_1(R_1R_2 + R_2R_3 + R_3R_1) - V_1(R_2R_3)}{R_1(R_1R_2 + R_2R_3 + R_3R_1}$$

$$IR_1' = \frac{V_1R_1R_2 + \cancel{V_1R_2R_3} + V_1R_3R_1 - \cancel{V_1R_2R_3}}{R_1(R_1R_2 + R_2R_3 + R_3R_1)}$$

$$IR_1' = \frac{\cancel{R_1}(V_1R_2 + V_1R_3)}{\cancel{R_1}(R_1R_2 + R_2R_3 + R_3R_1)}$$

$$\boxed{IR_1' = \frac{(V_1(R_2 + R_3))}{R_1R_2 + R_2R_3 + R_3R_1}} \quad \ldots (5)$$

case 2 :



apply KCL

$$\frac{VA}{R_1} + \frac{VA}{R_3} + \frac{VA - V_2}{R_2}$$

$$V_A\left(\frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1R_2R_3}\right) = \frac{V_2}{R_2}$$

$$V_A = \frac{V_2}{\cancel{R_2}}\left(\frac{R_1\cancel{R_2}R_3}{R_1R_2 + R_2R_3 + R_3R_1}\right)$$

$$\boxed{V_A = \frac{V_2R_1R_3}{R_1R_2 + R_2R_3 + R_3R_1}} \quad \ldots (6)$$

The response $I R_1'' = \frac{0 - V_A}{R_1}$

$$I R_1'' = \frac{0 - \frac{V_2 R_1 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1}}{R_1}$$

$$I R_1'' = \frac{- V_2 \cancel{R_1} R_3}{\cancel{R_1} (R_1 R_2 + R_2 R_3 + R_3 R_1)}$$

$$\boxed{I R_1'' = \frac{- V_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1}} \quad \cdots \text{(7)}$$

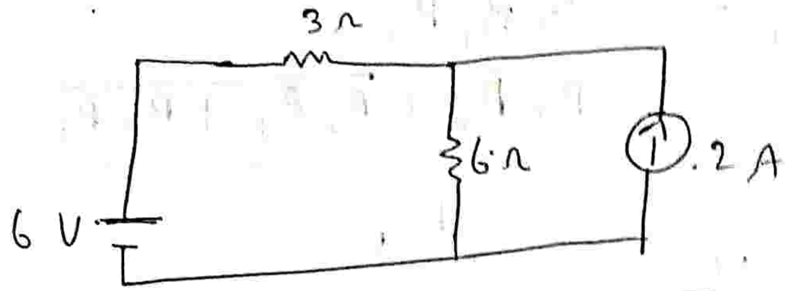
$$I R_1 = I R_1' + I R_1''$$

$$I R_1 = \frac{V_1 (R_2 + R_3)}{R_1 R_2 + R_2 R_3 + R_3 R_1} - \frac{V_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1}$$

$$\therefore \boxed{I R_1 = \frac{V_1 (R_2 + R_3) - V_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1}} \quad \cdots \text{(B)}$$

$\therefore A = B$ //

1) Find the current through 6 Ω resistance using super position theorem

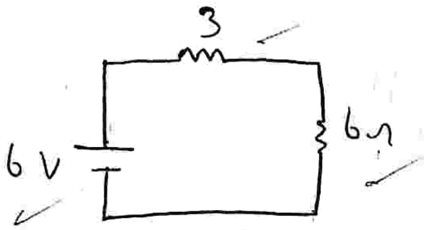


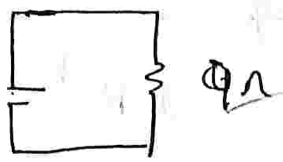
current through 6 Ω $\quad I_{6\Omega} = ?$

according to super position

$$I_{6\Omega} = I_{6\Omega}' + I_{6\Omega}''$$

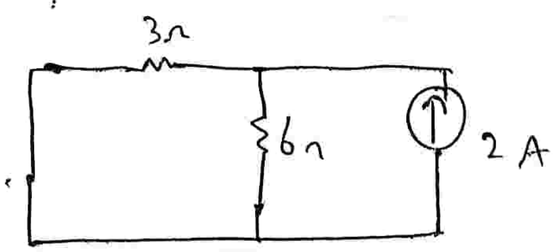
Case 1:
6 V acting alone





$$I_{6\Omega}' = \frac{6}{9} = 0.66\,A$$

Case 2:
2 A current source acting alone
short ckt 6 V (V) source.

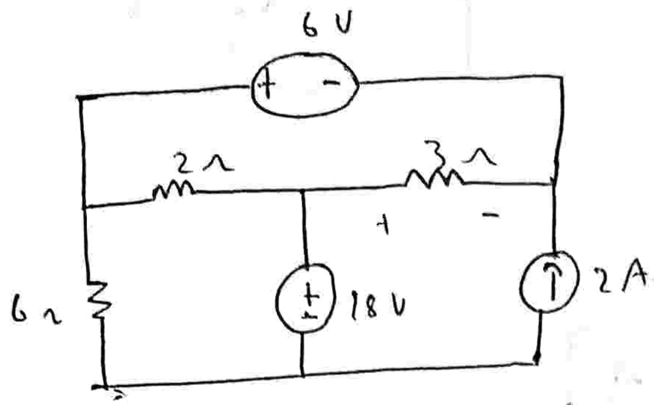


apply I D Rule.

$$I_{6\Omega}'' = \frac{2(3)}{6+3} = \frac{6}{9} = 0.66\,A$$

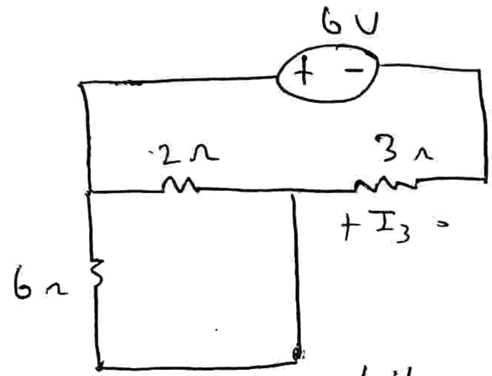
$I_{6\Omega} = 0.66\,A + 0.66\,A$ $\quad \boxed{I_{6\Omega} = 1.32\,A}$

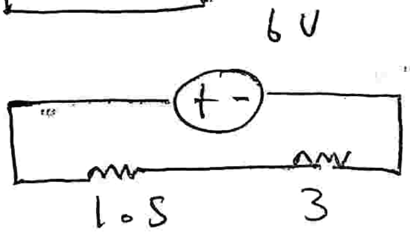
2) Find the voltage across 3 Ω resistor using super position theorem



soln: case ①

6 V acting alone. & 18V is shorted 2 A open





$$I_3 = \frac{6}{1.5+3}$$

$$\boxed{I_3 = 1.33 A}$$

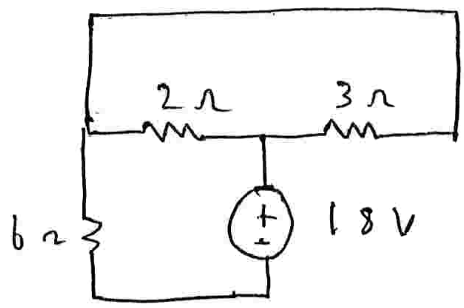
$$V' = IR$$

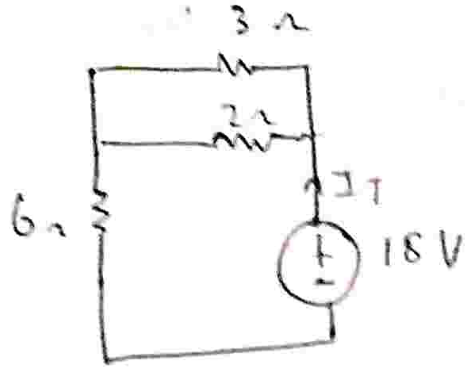
$$V' = 1.33 \times 3$$

$$\boxed{V' = 4 V}$$

case ②

consider 18 V alone short 6 V & open 2 A





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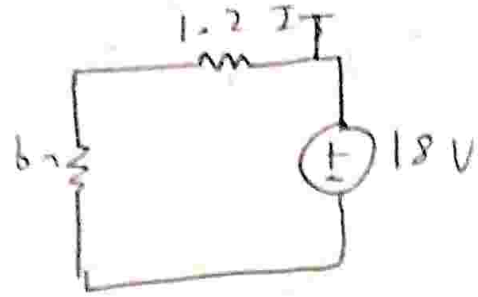


$$I_T = \frac{18}{7.2}$$

$$\boxed{I_T = 2.5A}$$

$$I_3 = \frac{I_T \times 2\Omega}{2+3}$$

$$I_3 = \frac{2.5 \times 2}{5}$$

$$\boxed{I_3 = 1A}$$

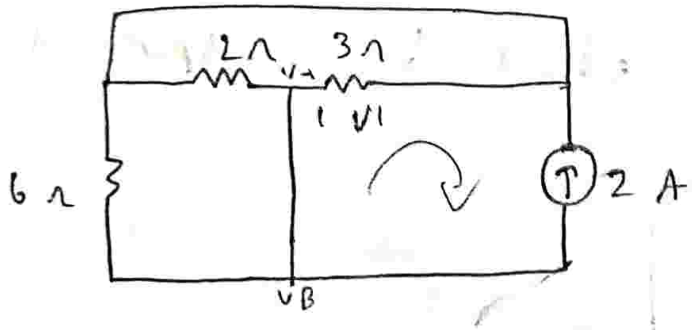
$V = IR$

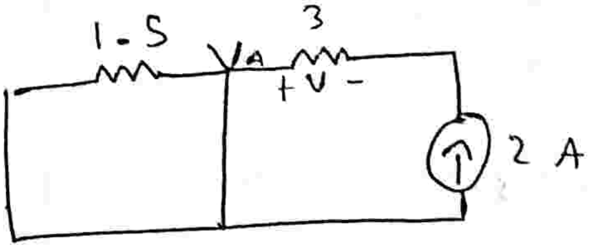
$V = 1 \times 3$

$$\boxed{V'' = 3V}$$

case ③

2A acting alone short 6V & 18V.





applying KCL

$$\frac{V_A - 0}{1.5} + \frac{V_A - 0}{3} = -2$$

$$V_A\left(\frac{1}{1.5} + \frac{1}{3}\right) = -2$$

$$\therefore V_A(1) = -2$$

$$\boxed{V_A = -2\ V}$$ or $$\boxed{V''' = -2\ V}$$

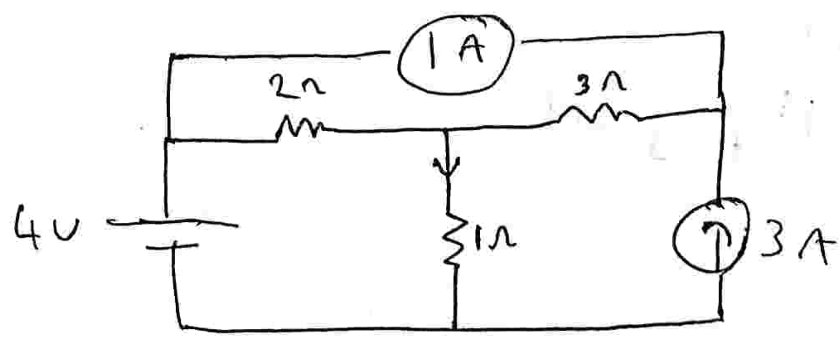
$$\therefore V_{3\Omega} = V_I + V_{II} + V_{III}$$

$$V_{3\Omega} = 4 + 3 - 2$$

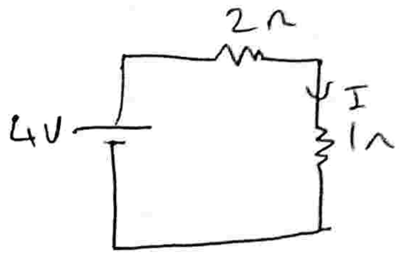
$$\boxed{V_{3\Omega} = 5\ V}$$

//

3) Find the current through 1Ω resistor using superposition theorem.



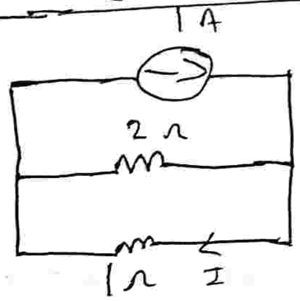
consider 4 V alone



$$I' = \frac{V}{R} = \frac{4}{2+1} = 1.33\,A$$

$$\boxed{I' = 1.33\,A}$$

Consider 1A current source: short (3 A current)

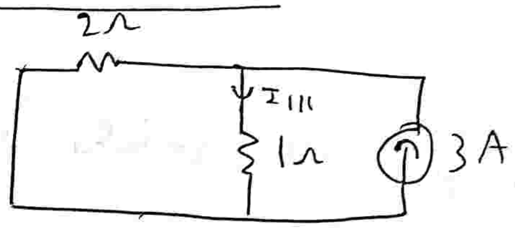


apply current divider

$$I''_{1\Omega} = \frac{1 \times (2)}{2+1} = 0.666$$

$$\boxed{I'' = 0.666A}$$

consider 3 A alone

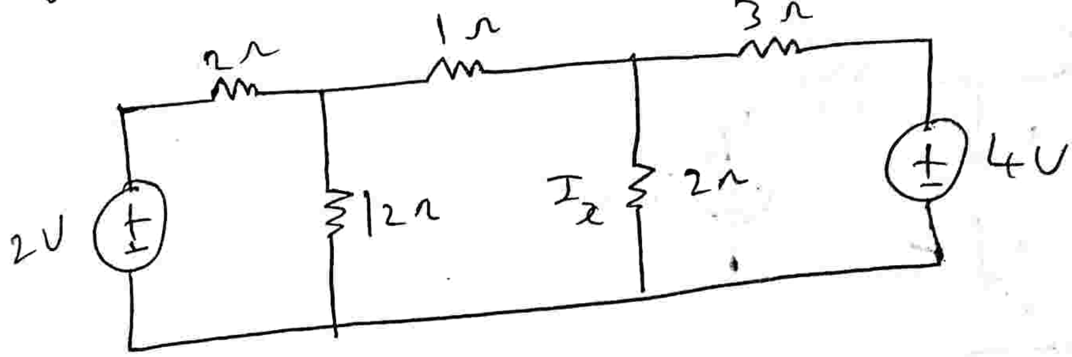


$$I''' = \frac{3\,(2)}{2+1} = 2A$$

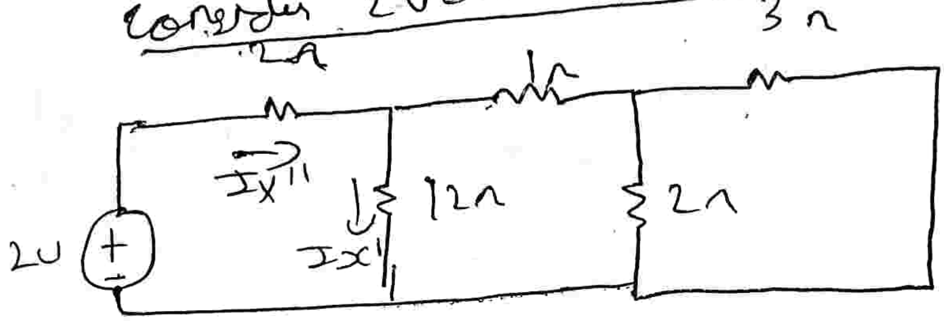
$$I = I' + I'' + I'''$$

$$\boxed{I = 4\,A}$$

4) Find the current in $R = 2\Omega$



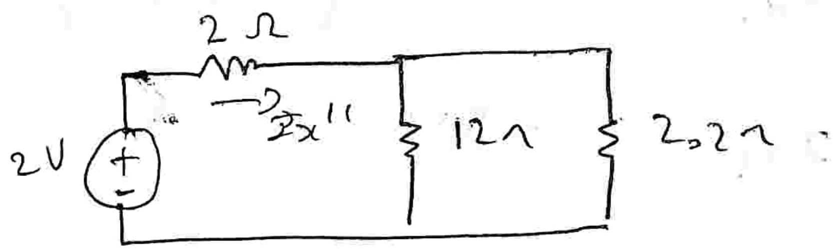
Consider 2V ~~alone~~ alone.



$3 \| 2 + 1\Omega$

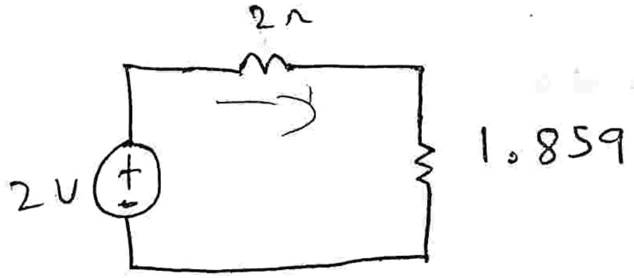
$0.2 + 1$

2.2Ω



$$I_x' = \frac{12 \times I_x''}{12 + 2.2}$$

$$\boxed{I_x' = 0.45 I_x}$$



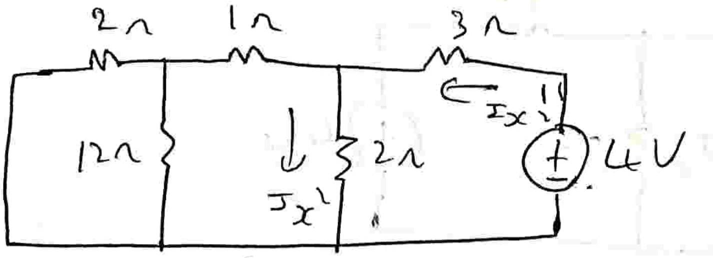
$$I_x'' = \frac{2}{1.859 + 2} = 0.5182\,A$$

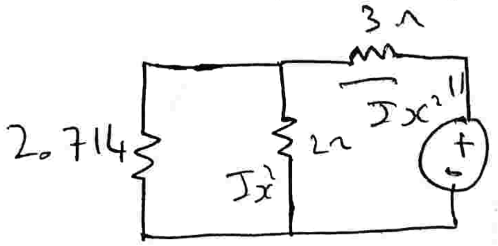
$$I_x' = 0.854 \times I_x''$$

$$\boxed{I_x = 0.854 \times 0.5182}$$

$$\boxed{\therefore I_x = 0.2627A}$$

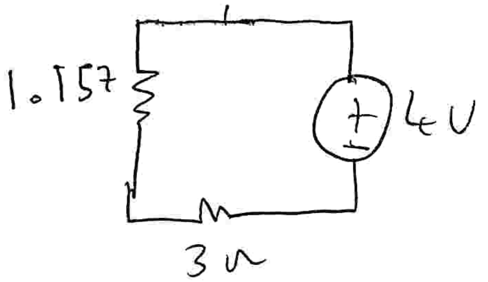
case (ii) let 4V be acting alone





$$I_{x_2} = \frac{2.714\, I_x''}{2+2.714}$$

$$\boxed{I_{x_2} = 0.572\, I_{x_2}'}$$



$$I_{x_2'} = \frac{V}{R} = \frac{4}{3+1.157}$$

$$I_{x_2} = 0.572\left(\frac{4}{3+1.157}\right)$$

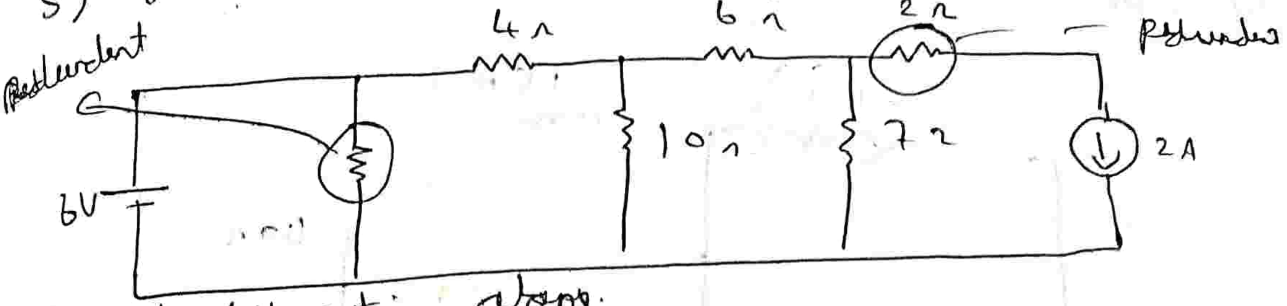
$$I_{x_2} = 0.0554$$

$$I_x = I_{x_1} + I_{x_2}$$

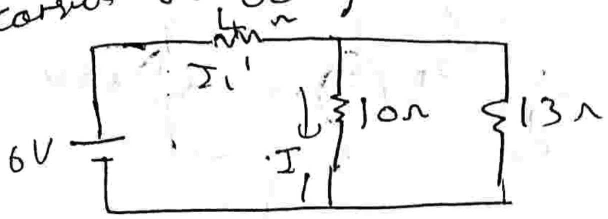
$$I_x = 0.2627 + 0.5540$$

$$\boxed{I_x = 0.8167\ A}$$

5) Find the current through 10 Ω resistor



consider 6 V acting alone.



$$I_1 = \frac{13 \times I_1'}{13+10}$$

$$\boxed{I_1 = 0.5652\, I_1'}$$

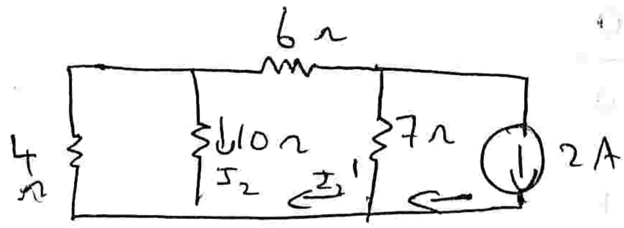
$$I_1' = \frac{6}{4 + 5.652}$$

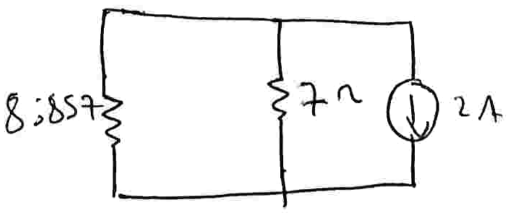
$$\boxed{I_1' = 0.6216}$$

$$I_1 = 0.5652\,(0.6216)$$

$$\boxed{I_1 = 0.35 A}$$

Case ② consider 2 A acting alone.





$$I_2 = -\frac{I_2' \times 4}{4+10}$$

$$I_2 = -0.285 I_2'$$

$$I_2' = \frac{7 \times 2}{8.857+7}$$

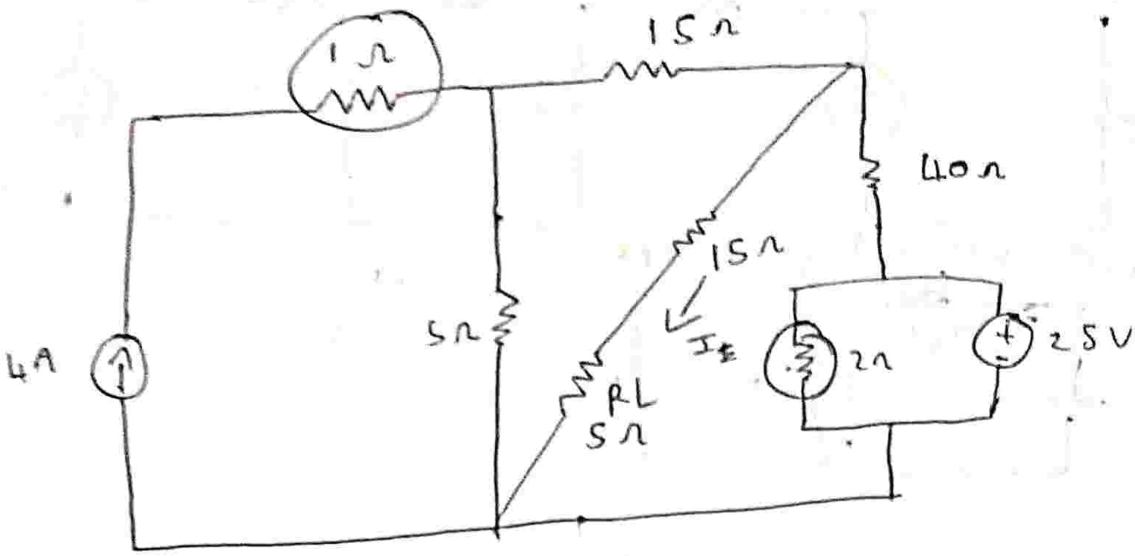
$$I_2' = 0.882 A$$

$$I_2 = -0.2857 \times 0.882$$

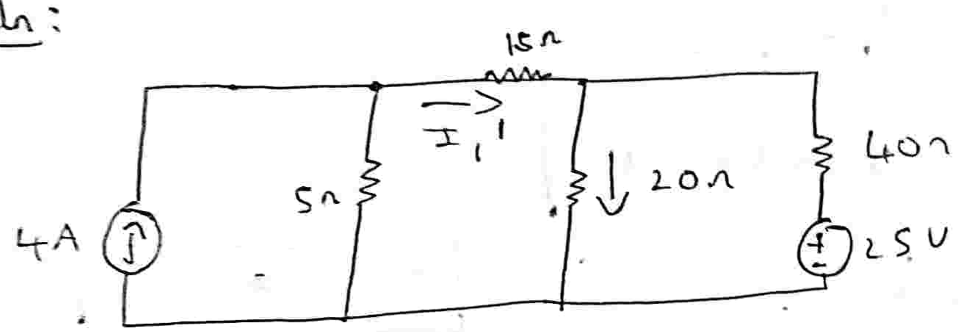
$$I_2 = -0.2857 \times 0.8825$$

$$\boxed{I = 0.099 A}$$

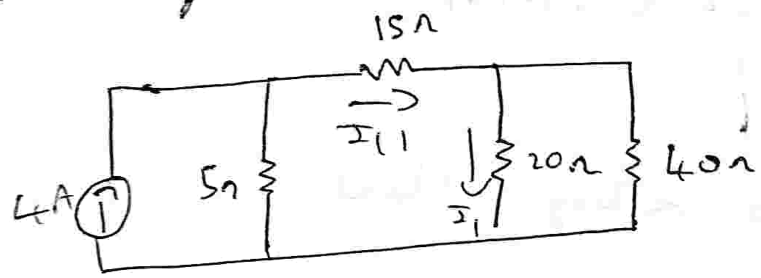
b] Find the current through RL



Soln:

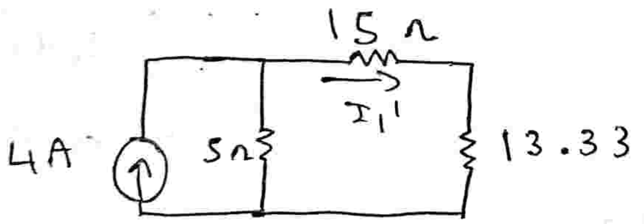


4A Acting alone



$$I_1 = \frac{I_1' \times 40}{40+20}$$

$$I_1 = 0.66 I_1'$$



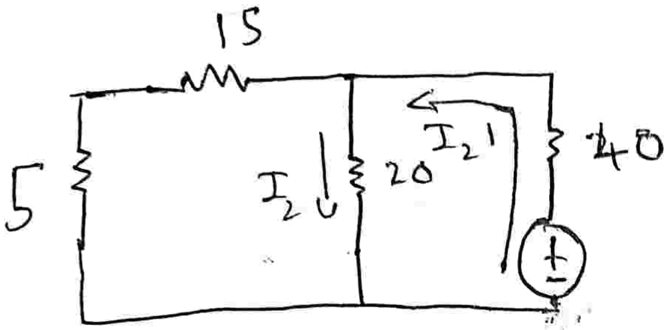
$$I_1' = \frac{4 \times 5}{5 + (15 * 13.33)}$$

$I_1' = 0.6A$

$I_1 = 0.6 \times 0.66$

$\boxed{I_1 = 0.4A}$

25V acting alone.



$$I_2 = \frac{I_2' \, 20}{20+20}$$

$\boxed{I_2 = 0.5 I_2'}$

$$I_2' = \frac{V}{R} = \frac{25}{40 + \left(\frac{20 \times 20}{40}\right)}$$

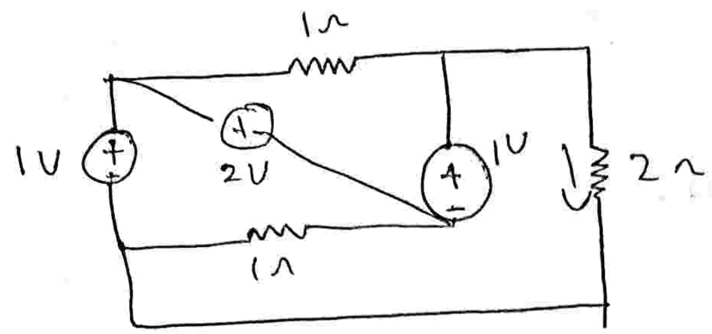
$\boxed{I_2' = 0.5A}$

$\boxed{I_2 = 0.25A}$

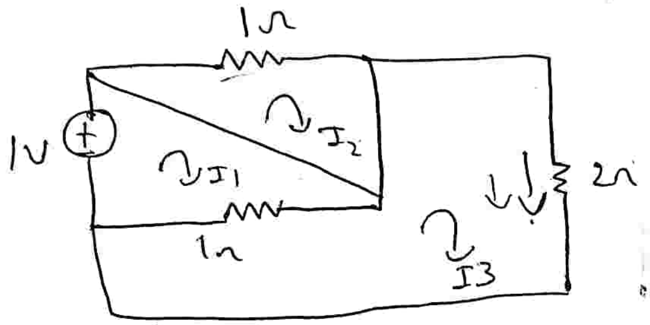
$I = 0.4 + 0.25$

$\boxed{I = 0.65A}$

7) Find the current through 2 Ω using super position theorem



case 1) consider 1V



apply KVL to loop ①

$$\boxed{-1I_1 + 1I_3 = -1} \quad \ldots ①$$

apply KVL to loop ②

$$\boxed{-1I_2 = 0} \quad \ldots ②$$

apply KVL to loop 3

$$-2I_3 - 1(I_3 - I_1) = 0$$

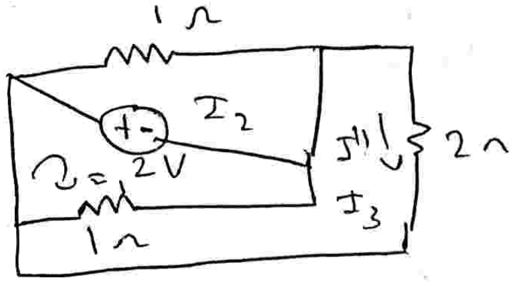
$$-2I_3 - I_3 + I_1 = 0$$

$$\boxed{I_1 - 3I_3 = 0} \quad \ldots ③$$

$I_1 = 1.5A$, $I_2 = 0$, $I_3 = 0.5A$

$$\boxed{I_3 = I_1 = 0.5A} \quad ..$$

case (ii) consider 2 V alone



loop ① KVL

$$-1I_1 + 1I_3 = 2 \quad \ldots ①$$

loop ② KVL,

$$-1I_2 = -2 \quad \ldots ②$$

loop ③ KVL

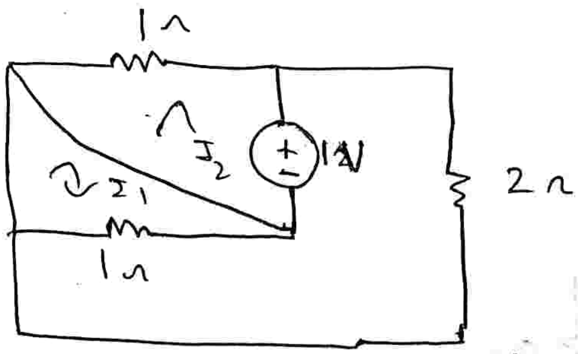
$$-2I_3 - I_3 + I_1 = 0$$

$$I_1 - 3I_3 = 0 \quad \ldots ③$$

$\boxed{I_1 = -3}$ $\boxed{I_2 = 2}$ $\boxed{I_3 = -1}$

$$\boxed{I_3 = I'' = -1A}$$

case (iii) consider 1V alone



apply KVL to loop ①

$$-1I_1 + 1I_3 = 0$$

apply KVL to loop ③

$$I_1 - 3I_3 = -1V$$

apply KVL to loop 2

$$-1I_2 = 1V$$

$\boxed{I_1 = 0.5A}$ $\boxed{I_2 = 1A}$ $\boxed{I_3 = 0.5A}$

$$\boxed{I_3 = I''' = 0.5 \ A}$$

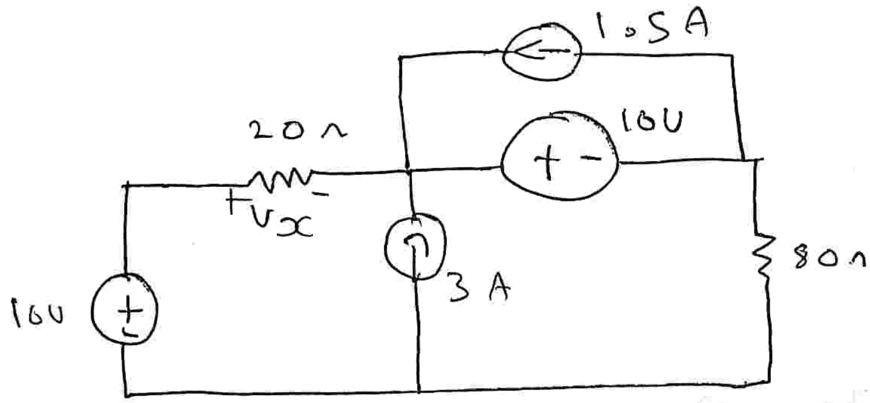
according to super position ~~theor~~ theorm

$$I = I' + I'' + I''$$

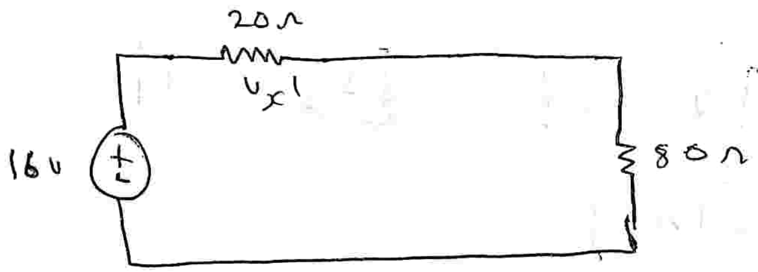
$$I = 0.5 - 1 + 0.5$$

$$\boxed{I = 0 A}$$

8) for the ckt shown find V_x



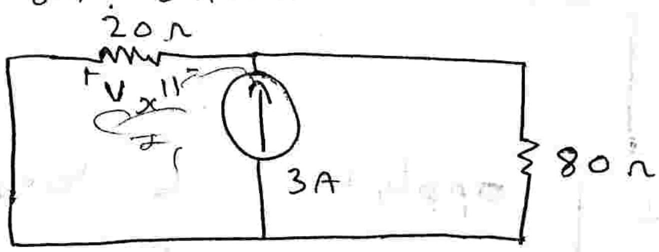
consider 16 V



$$V_x' = \frac{16 \times 20}{20+80}$$

$$\boxed{V_x' = 3.2 \ V}$$

consider 3 A alone.



apply current di.

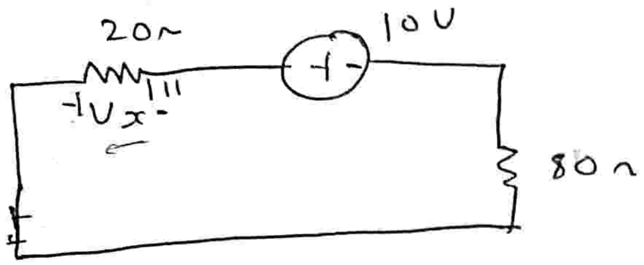
$I_{20} = \frac{3 \times 80}{20+30}$ $I_{20} = -2.4 A$

$V_x'' = I_{10} \times R$

$V_x'' = 2.4 \times 20$

$\boxed{V_x'' = -48\,V}$

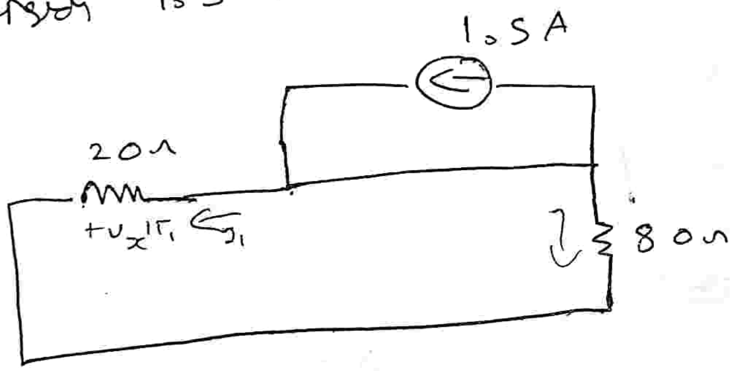
consider 10V alone.



$$V_x''' = \frac{10 \times 20}{80+20}$$

$\boxed{V_x''' = -2\,V}$

consider 1.5 A alone



as the is no resistor with current hence the I = 0

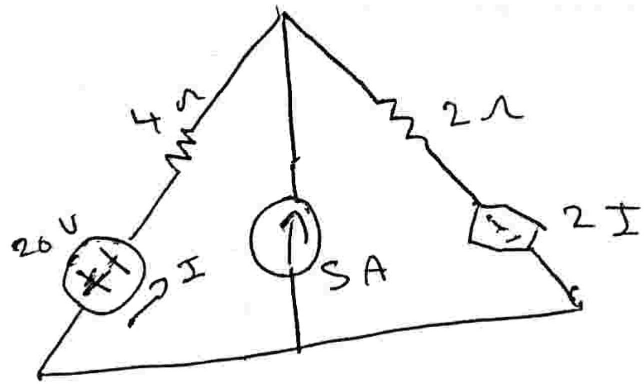
$\therefore \boxed{V_x'''' = 0\,V.}$

$\therefore V_x = V_x' + V_x'' + V_x''' + V_x''''$

$V_x = 3.2 - 2 + (-48)$

$\boxed{V_x = -46.8\,V}$

9)

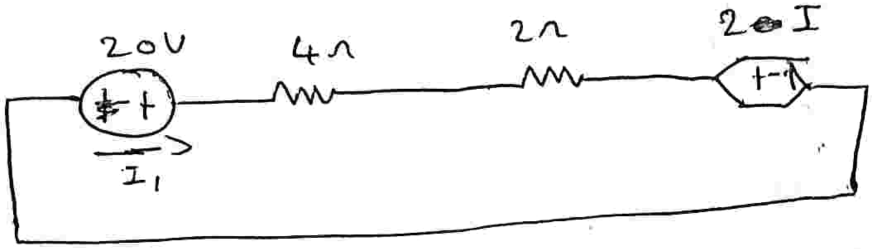


$I_1 = 2.5$

$I_2 = -1.25 A$

(In SPT if we have any dependent sources retain as it is in ckt)

Consider 20 V acting alone.



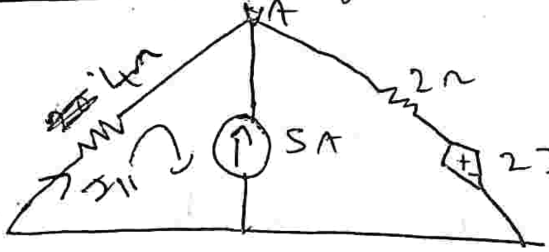
apply KVL

$20 - 4I - I2 - 2I = 0$

$20 - 8I = 0$

$$\boxed{I' = \cancel{\frac{20}{8}}\ 2.5 A}$$

Consider 5 A acting alone:



Apply KVL at loop ①

$-4 I_{11} - 5 = 0$

$-4 I_{11} = 5$

$I_{11} = \frac{-5}{4}$

$$\boxed{I'' = -1.25 \text{ A}}$$

→ Millman's Theorem:

consider the following network

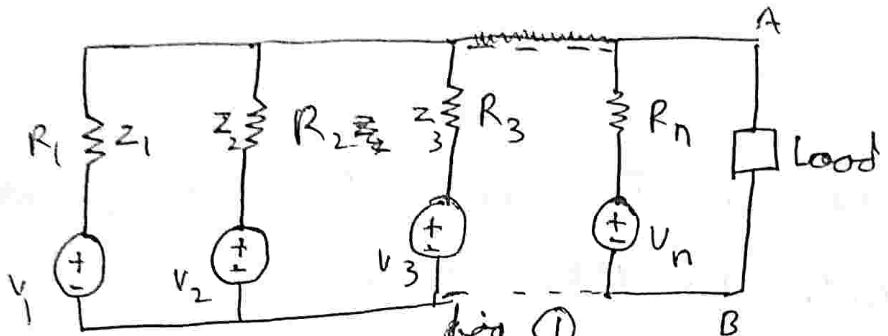


fig ①

Z = impedance (Ω) ohm

$Y = \frac{1}{Z}$ Admitance

In any linear Bilateral network consisting of of n number of voltage sources are converted into single voltage sources and all the impedancy are converted into single impedance across the load terminal.

The single converted voltage source is given by

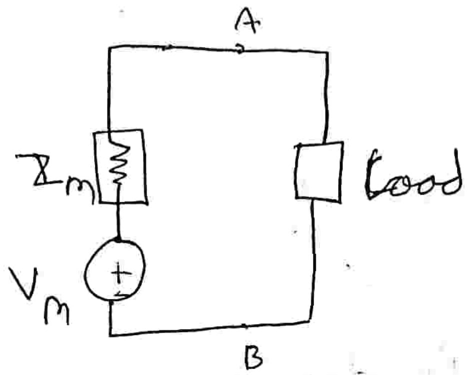
$$V_m = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3 + \dots + V_n Y_n}{Y_1 + Y_2 + Y_3 \dots + Y_n}$$

$$V_m = \frac{\sum_{K=1}^{n} V_K Y_K}{\sum Y_K}$$

$$Z_m = \frac{1}{Y_1 + Y_2 + Y_3 \dots + Y_n}$$

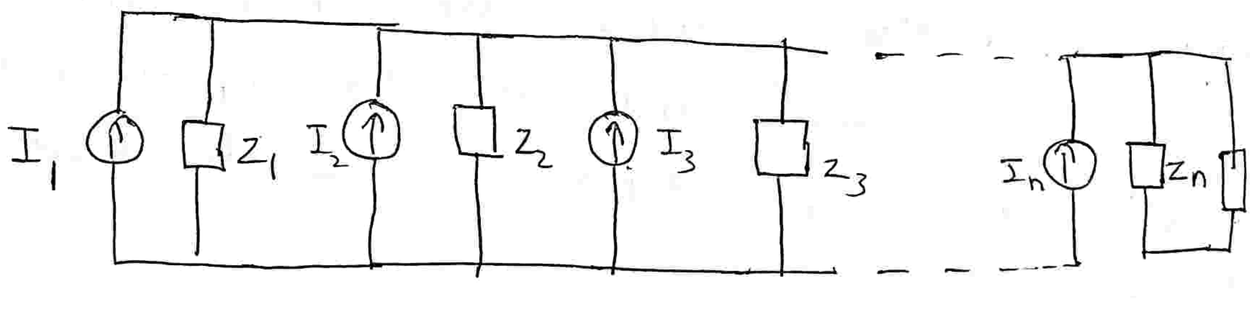
$$Z_m = \frac{1}{\sum_{K=1}^{n} Y_K}$$

millman's equivalent circuit is :



Proof : consider the fig ① consists of n no of voltage source connected in ||le across the load terminal A B

Convert all the voltage source into equivalent current source



$$I_1 = \frac{V_1}{Z_1} = V_1 Y_1$$

$$I_2 = \frac{V_2}{Z_2} = V_2 Y_2$$

$$I_3 = \frac{V_3}{Z_3} = V_3 Y_3$$

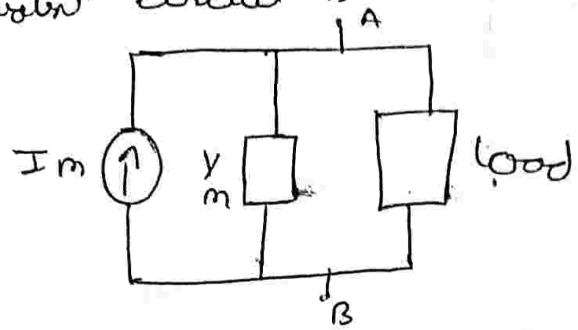
$$I_n = \frac{V_n}{Z_n} = V_n Y_n$$

we known that current sources and admitty in parallel can be added.

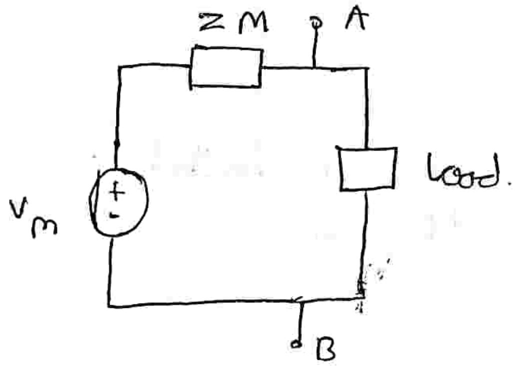
$$I_1 + I_2 + I_3 \cdots\cdots I_n = I_m$$

$$Y_1 + Y_2 + Y_3 \cdots Y_n = Y_m$$

The equivalent circuit is



convert above current source into equivalent voltage source.



$$V = IR$$

$$V_m = I_m Z_m$$

$$V_m = \frac{I_m}{Y_m} \quad \cdots \text{(1)}$$

$$V_m = \frac{I_1 + I_2 + I_3 \cdots + I_n}{y_1 + y_2 + y_3 \cdots + y_n} \quad \cdots \text{(2)}$$

WKT

$$V = IZ$$

$$I = \frac{V}{Z} = V\cancel{Y} \quad \cdots \text{(3)}$$

sub ③ in ②

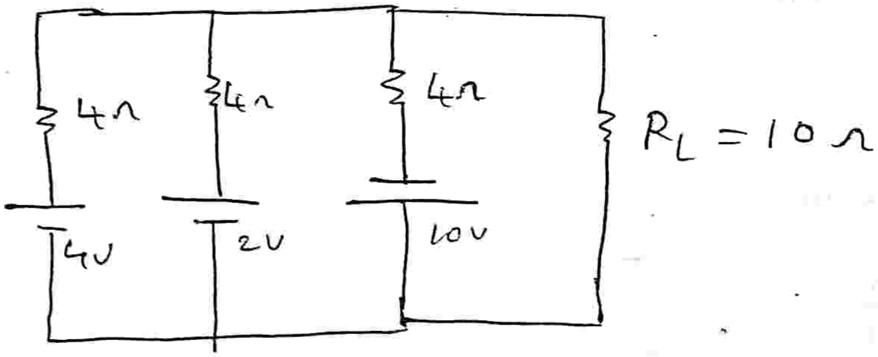
$$V_m = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3 \cdots + V_n Y_n}{Y_1 + Y_2 + Y_3 \cdots + Y_n}$$

$$\boxed{V_m = \frac{\sum_{K=1}^{n} V_K Y_K}{\sum_{K=1}^{n} Y_K}}$$

$$Z_m = \frac{1}{Y_1 + Y_2 + Y_3 \cdots + Y_n}$$

$$\boxed{Z_m = \frac{1}{\sum_{K=1}^{n} Y_K}}$$

① Find the value of current in loaded resistance $R_L = 10\,\Omega$ using millmans theorem



Soln. $Z_1 = 4 \quad Z_2 = 4 \quad Z_3 = 4$

$$Y_1 = Y_2 = Y_3 = \frac{1}{4} = 0.25$$

$$V_1 = 4 \quad V_2 = 2 \quad V_3 = -10$$

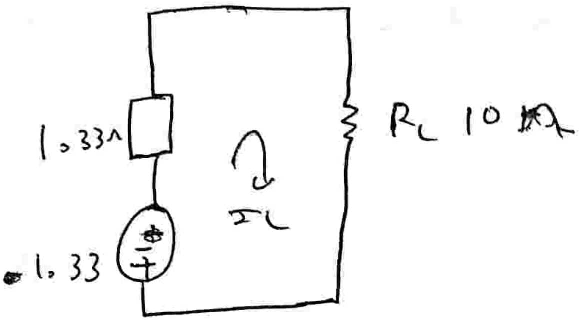
$$V_m = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3}{Y_1 + Y_2 + Y_3}$$

$$V_m = \frac{(4 \times 0.25) + (2 \times 0.25) + (-10 \times 0.25)}{0.75}$$

$$\boxed{V_m = -1.33\,V}$$

$$Z_m = \frac{1}{0.75}$$

$$\boxed{Z_m = 1.33\,\Omega}$$

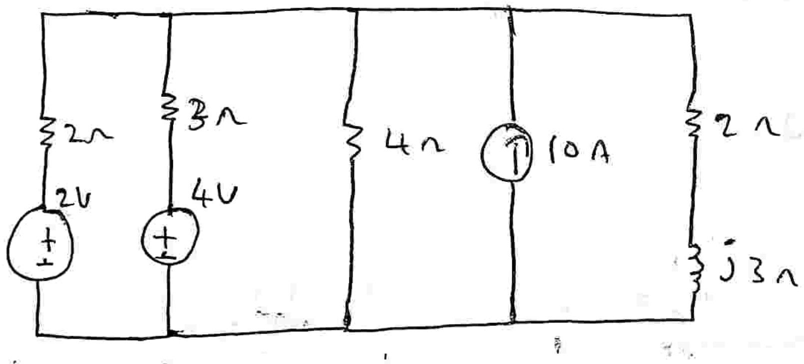


$$I_L = \frac{1.33}{1.33 \times 10}$$

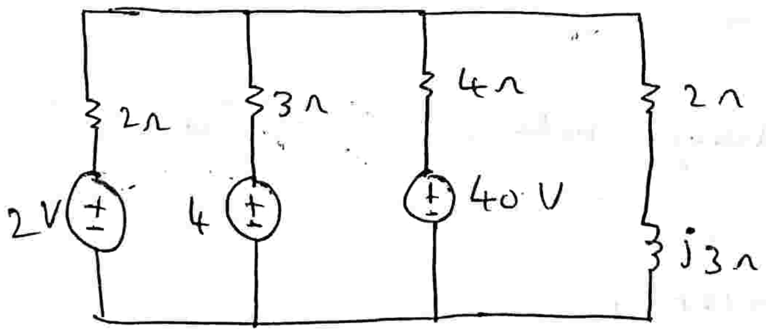
$$\boxed{I_L = 0.117\,A}$$

---

(2)



Find the current $Z_L = 2 + j3\,\Omega$



$Z_1 = 2$ $\qquad Z_2 = 3$ $\qquad Z_3 = 4$

$Y_1 = \frac{1}{2} = 0.5$ $\qquad Y_2 = \frac{1}{3} = 0.33$ $\qquad Y_3 = \frac{1}{4} = 0.25$

$\boxed{V_1 = 2V}$ $\boxed{V_2 = 4V}$ $\boxed{V_3 = 40V}$

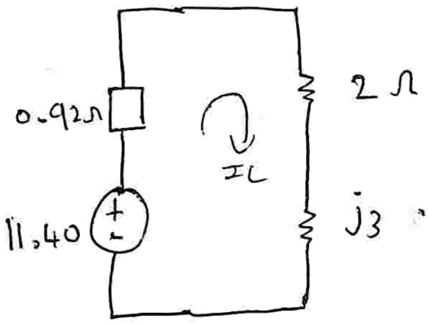
$$V_m = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3}{Y_1 + Y_2 + Y_3}$$

$$V_m = \frac{(2 \times 0.5) + (4 \times 0.33) + (0.25 \times 40)}{0.5 + 0.33 + 0.25}$$

$$\boxed{V_m = 11.40V}$$

$$Z_m = \frac{1}{Y_1 + Y_2 + Y_3} = \frac{1}{0.5 + 0.33 + 0.25}$$

$$\boxed{Z_m = 0.92\Omega}$$

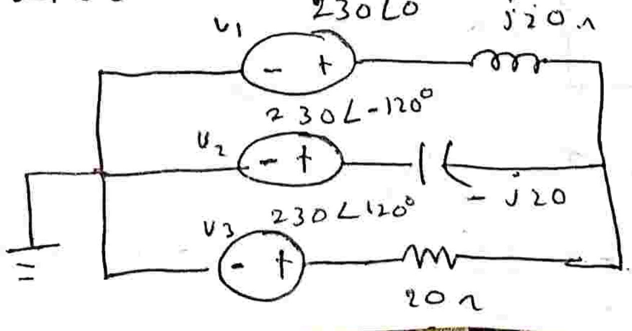


$$I_L = \frac{V}{R}$$ $$\boxed{I_L = 1.899 - 1.951jA}$$

or

$$\boxed{I_L = 2.72 \angle -45.77 A}$$

③. convert the following network into equivalent millman's circuit



$Z_1 = j20$ $\quad Z_2 = -j20$ $\quad Z_3 = 20$

$\boxed{Y_1 = \frac{1}{j20} = -0.05j}$ $\quad \boxed{Y_2 = \frac{1}{-j20} = 0.05j}$ $\quad Y_3 = \frac{1}{20}$

$\boxed{Y_3 = 0.05}$

$V_1 = 230 \angle 0 = 230$

$V_2 = 230 \cancel{\angle -120^\circ} \angle -120^\circ$

$\boxed{V_2 = -115 - 199.18i}$

$V_3 = 20 \angle 120$

$\boxed{V_3 = -115 + 199.18i}$

$$Z_m = \frac{1}{Y_1 + Y_2 + Y_3}$$

$$Z_m = \frac{1}{-0.05j + 0.05i + 0.05}$$

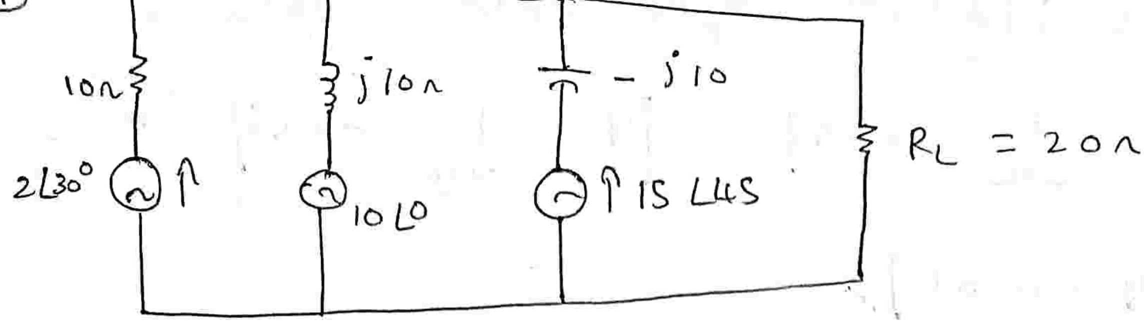
$\boxed{Z_m = 20\Omega}$

$$V_m = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3}{Y_1 + Y_2 + Y_3}$$

-115
$+99.1$

$$V_m = \frac{230(-0.05i) + ((-115 - 199.18i)(0.05i)) + ((0.05$$
$$-0.05j + 0.05i + 0.05$$

$\boxed{V_m = 84.18 - 145.82i\ V}$ or $\boxed{V_m = 168.37 \angle 60^\circ V}$



$Z_1 = 10 \qquad Z_2 = j10 \qquad Z_3 = -j10$

$\boxed{Y_1 = \frac{1}{Z_1} = 0.1} \qquad \boxed{Y_2 = \frac{1}{j10} = -0.01i} \qquad \boxed{Y_3 = \frac{1}{-j10} = 0.1i}$

$V_1 = 2\angle 30^\circ \quad \boxed{V_1 = 1.73 + i}$

$\boxed{V_2 = 10}$

$V_3 = 15\angle 45 \quad \boxed{V_3 = 10.60 + 10.60i}$

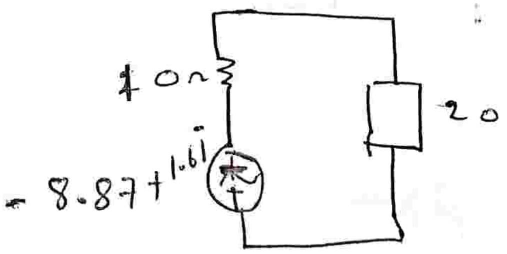
$$Z_m = \frac{1}{0.1 + (-0.1i) + (0.1i)}$$

$\boxed{Z_m = 10\,\Omega}$

$$V_m = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3}{Y_1 + Y_2 + Y_3}$$

$$V_m = \frac{(1.73 + i)(0.1) + (10)(-0.01i) + (10.60 + 10.60i)(0.1i)}{0.1 + (-0.1i) + (0.1i)}$$

$\boxed{V_m = -8.87 + 1.6i \ \text{V}}$



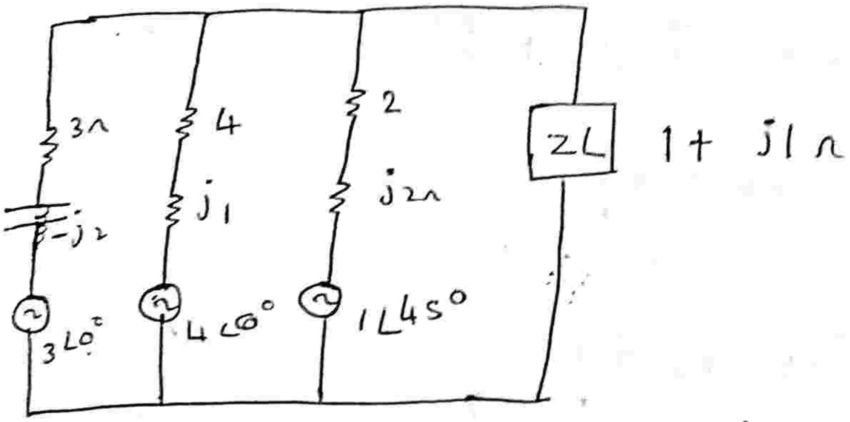
$\boxed{I_L = 0.3\angle 169.77\ \text{A}}$

or

$I_L = \frac{8.87 + 1.6i}{20 + 10}$

$\boxed{I_L = -0.295 + 0.051i\ \text{A}}$

5. Find the current through to ZL



$z_1 = 3 - j2$ $\quad Z_2 = 4 + j1$ $\quad Z_3 = 2 + j2\Omega$

$Y_1 = \frac{1}{Z_1} = \frac{1}{3-j2}$ $\quad Y_2 = \frac{1}{4+j1}$ $\quad y_3 = \frac{1}{2+j2}$

$\boxed{Y_1 = 0.23 + 0.15i}$ $\quad \boxed{Y_2 = 0.23 - 0.05i}$ $\quad \boxed{y_3 = 0.25 - 0.25i}$

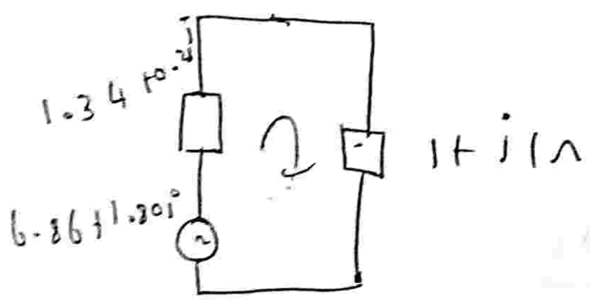
$$Zm = \frac{1}{Y_1 + Y_2 + Y_3}$$

$$Zm = \frac{1}{(0.23 + 0.15i) + (0.23 - 0.05i) + (0.25 - 0.25i)}$$

$$\boxed{Zm = 1.34 + 0.28i}$$

$\boxed{V_1 = 3}$ $\quad \boxed{V_2 = 4}$ $\quad \boxed{V_3 = 0.707 + 0.707i}$

$$\boxed{V_m = 6.86 + 1.80i \ V}$$



$$I_L = \frac{6.86 + 1.80i}{(1.34 + 0.28i) + (1 + 1i)}$$

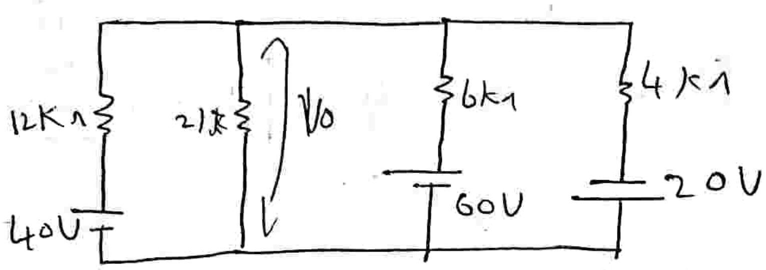
$$\boxed{I_L = 2.58 - 0.64i \text{ A}}$$

or

$$\boxed{I_L = 2.65 \angle -13.97 \text{ A}}$$

---

6)



$Z_1 = 12K\Omega$ $\quad Z_2 = 6K$ $\quad Z_3 = 4k$

$$\boxed{Y_1 = \frac{1}{12 \times 10^3} = 8.33 \times 10^{-5}} \quad \boxed{Y_2 = \frac{1}{6 \times 10^3} = 1.66 \times 10^{-4}}$$

$$\boxed{Y_3 = \frac{1}{4K} = 2.5 \times 10^{-4}}$$

$$Z_m = \frac{1}{Y_1 + Y_2 + Y_3}$$

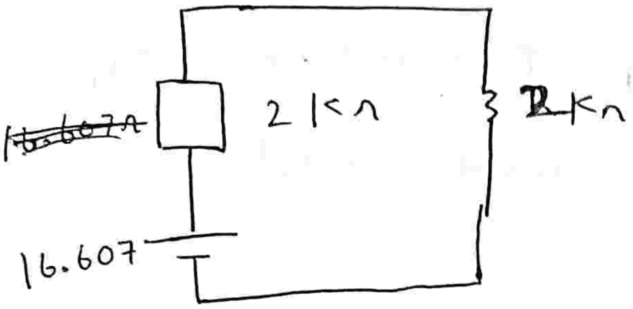
$$Z_m = \frac{1}{8.33 \times 10^{-5} + 2.5 \times 10^{-4} + 1.66 \times 10^{-4}}$$

$$\boxed{Z_m = 2 K\Omega}$$

$$V_m = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3}{Y_1 + Y_2 + Y_3}$$

$$V_m = \frac{(40 \times 8.33 \times 10^{-5}) + (60 \times 1.66 \times 10^{-4}) + (-20 \times 2.5 \times 10^{-4})}{8.33 \times 10^{-5} + 1.66 \times 10^{-4} + 2.5 \times 10^{-4}}$$

$$\boxed{V_m = 16.607\,\Omega}$$



$$V_{Th} = \frac{16.607 \times 2 \times 10^3}{2 \times 10^3 + 2 \times 10^3}$$

$$\boxed{V_o = 8.3\ V}$$

Thevenin's Theorem:

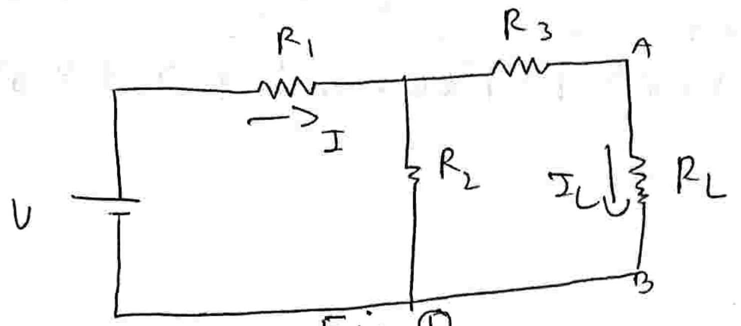


Fig ①

In any linear Bilateral network is represents a single voltage ~~state~~ source V_{th} & Resistance R_{th} and current through the load resistance is given by. $\boxed{I_L = \frac{V_{th}}{R_{th} + R_L}}$.

Thevenin's Voltage is obtained by load resistance is removed & R_{th} is obtained by removing the load as well as if there is any voltage source make it short ckt, if any current source open ckt (open the currents)

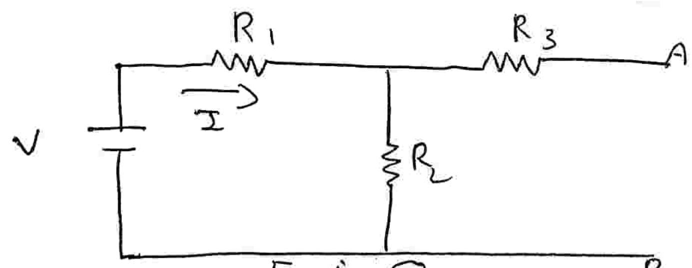


Fig ②

$$V_{th} = I \cdot R_2$$

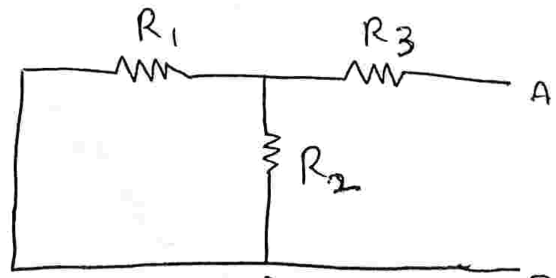


Fig ③

$$R_{th} = (R_1 \| R_2) + R_3$$

Now consider the n/w shown in fig ①

from ckt

$$\boxed{I_L = \frac{I R_2}{R_2 + (R_3 + R_L)}} \quad \ldots ①$$

$$I = \frac{V}{(R_3 + R_L \| R_2) + R_1}$$

$$I = \frac{V}{R_1 + \frac{R_2 (R_3 + R_L)}{R_2 + R_3 + R_L}}$$

$$I = \frac{V}{\frac{R_1 R_2 + R_1 R_3 + R_1 R_L + R_2 R_3 + R_3 R_L}{R_2 + R_3 + R_L}}$$

$$\boxed{I = \frac{V (R_2 + R_3 + R_L)}{R_1 R_2 + R_1 R_3 + R_1 R_L + R_2 R_3 + R_3 R_L}} \quad \ldots ②$$

subs ② in ①

$$I_L = \frac{\frac{V(\cancel{R_2} + \cancel{R_3} + \cancel{R_L})}{R_1 R_2 + R_1 R_3 + R_1 R_L + R_2 R_3 + R_3 R_L} \cdot R_2}{\cancel{R_2} + (\cancel{R_3} + \cancel{R_L})}$$

$$I_L = \frac{V R_2}{R_1 R_2 + R_1 R_3 + R_1 R_L + R_2 R_3 + R_3 R_L} \quad \ldots ③$$

To prove :

To find VTH,

consider fig ②

$$V_{th} = I R_2 \quad \ldots \text{④}$$

To find Rth

consider fig ③

$$R_{th} = (R_1 \parallel R_2) + R_3$$

$$R_{th} = \frac{R_1 R_2}{R_1 + R_2} + R_3.$$

$$R_{th} = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1 + R_2} \quad \ldots \text{⑤}$$

The current through load resistance is given by.

$$I_L = \frac{V_{th}}{R_{th} + R_L} \quad \ldots \text{⑥}$$

$$I_L = \frac{I R_2}{\left(\frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1 + R_2}\right) + R_L}$$

$$I_L = \frac{I R_2 (R_1 + R_2)}{R_1 R_2 + R_1 R_3 + R_2 R_3 + R_1 R_L + R_2 R_L}$$

$$I = \frac{V}{R_1 + R_2} \quad \ldots \text{(4)}$$

$$I_L = \frac{\frac{V}{\cancel{R_1+R_2}} R_2 \cancel{(R_1+R_2)}}{R_1 R_2 + R_1 R_3 + R_2 R_3 + R_1 R_L + R_2 R_L}$$

$$I_L = \frac{V R_2}{R_1 R_2 + R_1 R_3 + R_2 R_3 + R_1 R_L + R_2 R_L} \quad \ldots \text{(8)}$$

eq (3) = eq (8)

$\boxed{\text{Hence proved}}$ //

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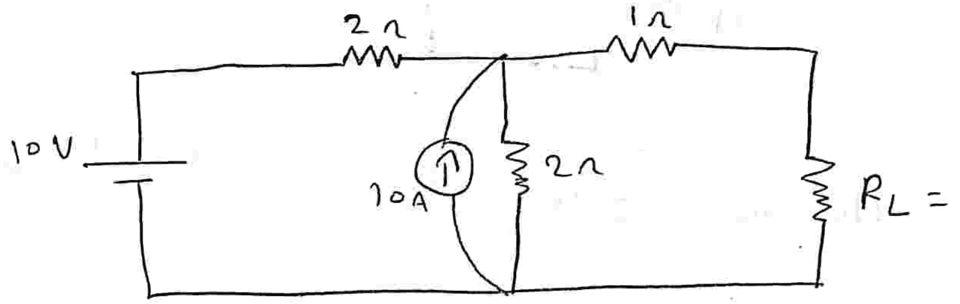
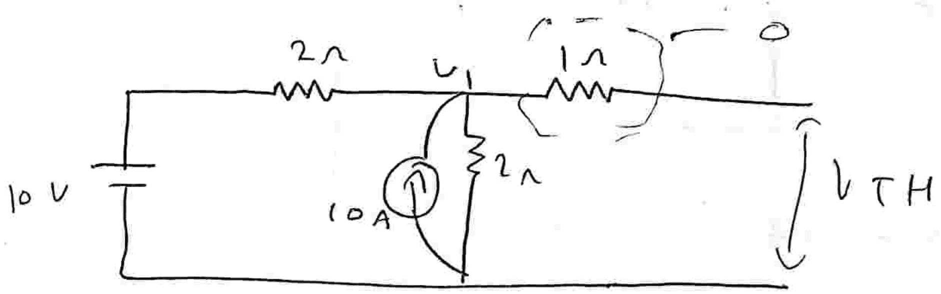
1) using thevenin's theorem Find the load current



To find V_{TH}



$$V_{TH} = V_1$$

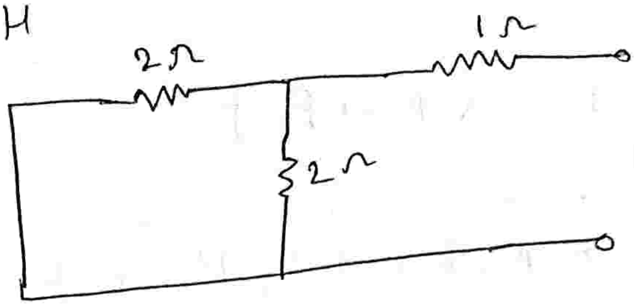
$$\frac{V_1 - 10}{2} + \frac{V_1}{2} = 10$$

$$\frac{V_1}{2} + \frac{V_1}{2} = 10 + 5$$

$$\boxed{V_1 (1) = 15}$$

$\boxed{V_{TH} = 15\,V}$

To find R_{TH}

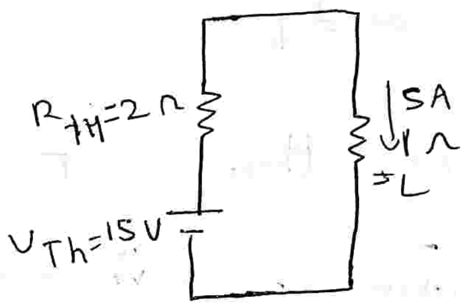


$R_{TH} = (2 \| 2) + 1$

$\boxed{R_{TH} = 2\,\Omega}$

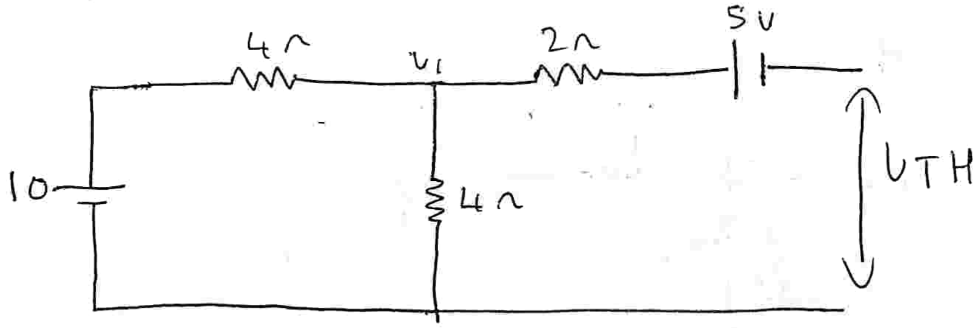
$$I_L = \frac{V_{TH}}{R_{TH} + R_L}$$

$$I_L = \frac{15}{2+1}$$



$\boxed{I_L = 5A}$

2) using thevinings theorem find the current through RL



$$\frac{V_1 - 10}{4} + \frac{V_1 \cancel{-5}}{2} + \frac{V_1}{4} =$$

$V_{TH} = V_1 - 5$

$$\frac{V_1}{4} - \frac{10}{4} + \frac{V_1}{4} = 10 \cancel{2 - 5}$$

$V_1 = 15$

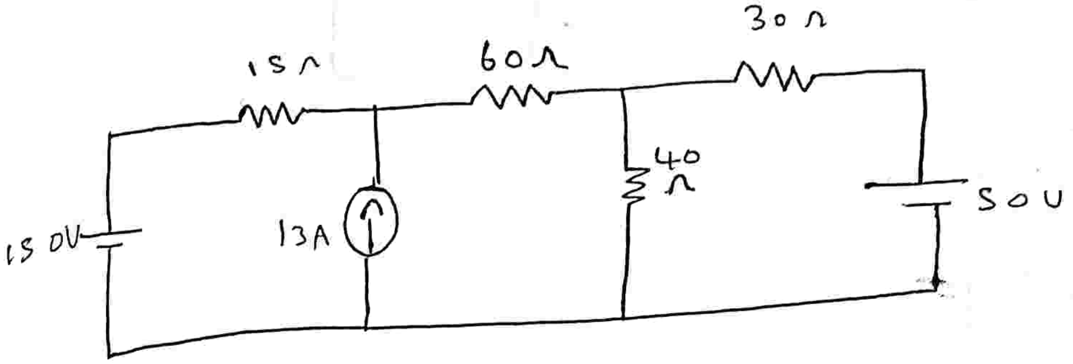
$V_{Th} = 15 - 5$

$\boxed{V_{Th} = 10V}$

$R_{Th} = (2 \| 2) + 2$

$\boxed{R_{th} = 4\Omega}$

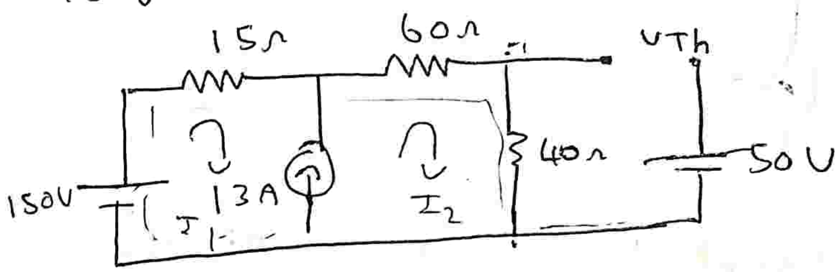
3) using thevenins theorem find the current through 30Ω resistor.



Soln:

$R_L = 30\Omega$

To find V_{TH} open $R_L = 30\Omega$



$I_2 - I_1 = 13 \quad \ldots \quad (1)$

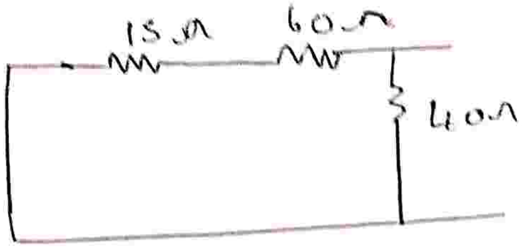
apply KVL to super mesh

$150 - 15I_1 - 60I_2 - 40I_2 = 0$

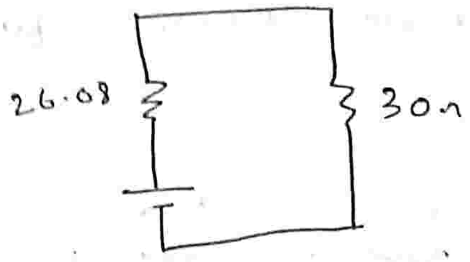
$-15I_1 - 100I_2 = -150$

$\boxed{I_1 = -10A}$ $\boxed{I_2 = 3}$

$V_{Th} = 70V$



$$\boxed{R_{TH} = 26.08\,\Omega}$$



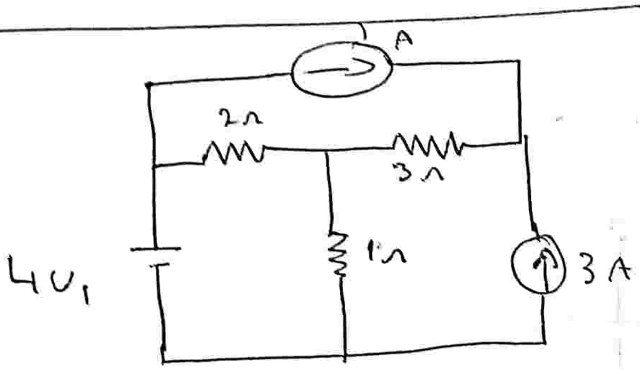
$$I_L = \frac{V_{Th}}{R_{Th}+R_L}$$

$$I_L = \frac{70}{26.08+30}$$

$$\boxed{I_L \quad 1.24A}$$

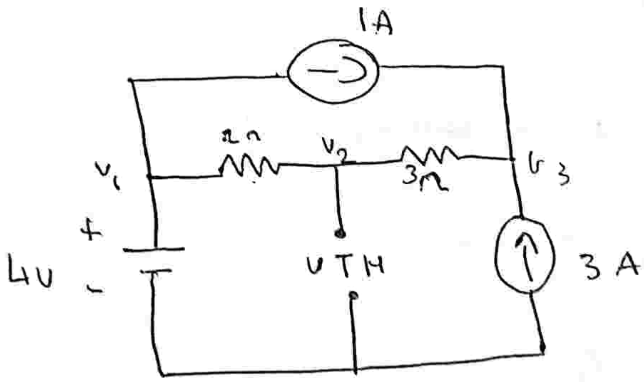
---

4)



Find the current through 1Ω using thevenins theorem

soln: Remove R_L 1Ω



From the ckt $\boxed{V_1 = 4V}$

$$\boxed{V_2 = V_{TH}}$$

$$\frac{V_2 - 4}{2} + \frac{V_2 - V_3}{3} = 0$$

$$V_2\left(\frac{1}{2} + \frac{1}{3}\right) - V_3\left(\frac{1}{3}\right) = 2$$

$$0.833\,V_2 - 0.33\,V_3 = 2$$

apply KCL to node 3

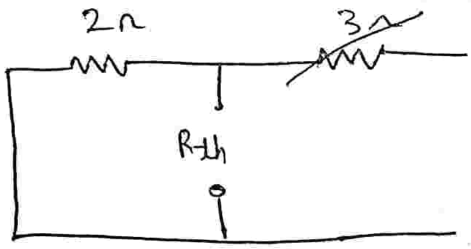
$$\frac{V_3 - V_2}{3} = 1 + 3$$

$$-0.33\,V_2 + 0.33\,V_3 = 4$$

$$V_2 = 12\text{ V}$$

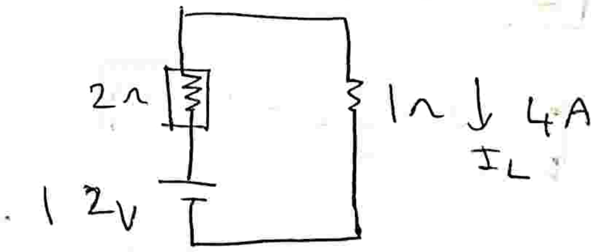
$$\therefore V_{TH} = 12\text{V}$$

$R_{TH} =$



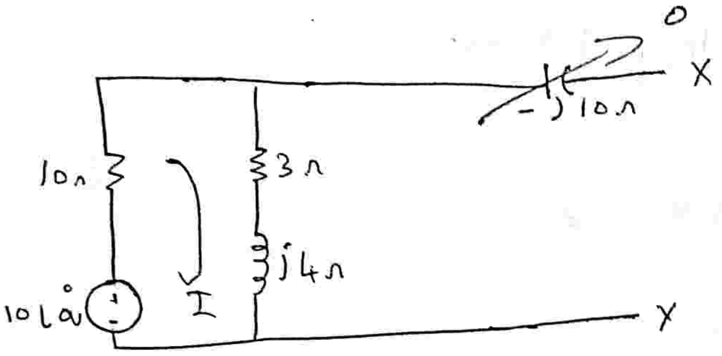
$$R_{TH} = 2\Omega$$

$$I_L = \frac{12}{2+1}$$



$$\boxed{I_L = 4\text{A}}$$

5) obtain the thevenin equivalent for the ckt shown across the terminal x and y



To find VTH open load.

$$V_{TH} = I(3+j4)$$

apply KVL to loop 1

$$10 - 10I - (3 + j4)I = 0$$

$$10 + (-13 - 4i)I = 0$$

$$\boxed{I = \frac{-10}{-13-4i}}$$

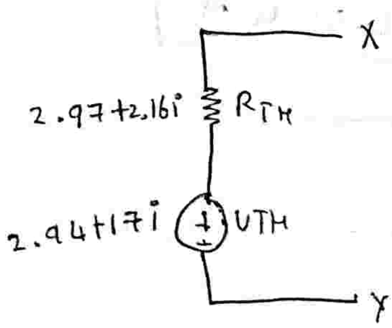
$$\boxed{I = 0.70 - 0.21i}$$

$$\boxed{I = 0.73 \angle -17.10}$$

$$V_{TH} = 0.70 - 0.21i(3+j4)$$

$$\boxed{V_{TH} = 3.654 \angle 36.43}$$

$$\boxed{V_{TH} = 2.94 + 2.17i}$$



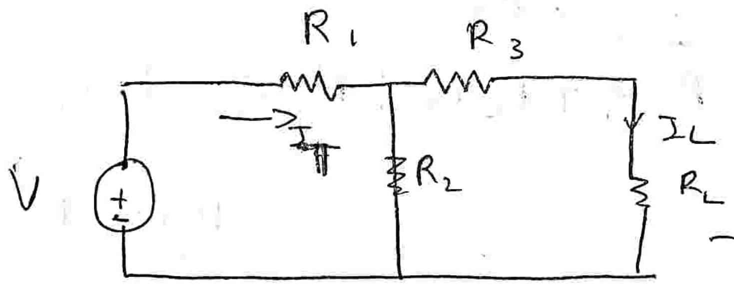
$R_{TH} = 10 \,||\, (3 + j4)$

$$\boxed{R_{TH} = 2.97 + 2.16\,i}$$

→ Norton's theorem :

In any linear bilateral network, however the network may be complicated, the n/w may be replaced by single current source in parallel with an impedence.

The current of current source is equal to the short circuit current flowing through the load terminal and the value of Z is equivalent to the impedence of the n/w ~~for~~ viewed from the load terminal and replace all independent voltage source by short ckt and current by open ckt.



→ fig ①

consider the following ckt

from the ckt

$$I_L = \frac{I_T\, R_2}{R_2 + R_L + R_3} \quad \ldots\ldots ①$$

$$I_T = \frac{V}{R_T} \quad \ldots ②$$

$R_T = (R_3 + R_L \,||\, R_2) + R_1$

$$R_T = R_1 + \frac{R_2 (R_3 + R_L)}{R_2 + R_3 + R_L}$$

$$R_T = \frac{R_1 R_2 + R_3 R_1 + R_L R_1 + R_2 R_3 + R_2 R_L}{R_2 + R_3 + R_L}$$

subs ③ in ②

$$I_T = \frac{V (R_2 + R_3 + R_L)}{R_1 R_2 + R_3 R_1 + R_L R_1 + R_2 R_3 + R_2 R_L} \quad \cdots ④$$

subs 4 in ①

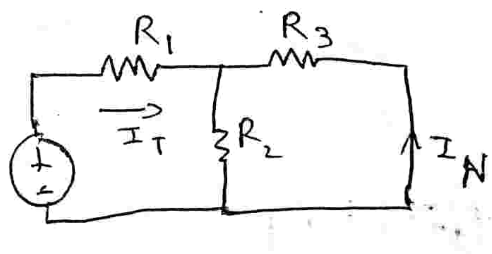
$$I_L = \frac{\dfrac{V \cancel{(R_2 + R_3 + R_L)}}{R_1 R_2 + R_3 R_1 + R_L R_1 + R_2 R_3 + R_2 R_L}}{\cancel{R_3 + R_2 + R_L}} \times R_2$$

$$I_L = \frac{V R_2}{R_1 R_2 + R_3 R_1 + R_L R_1 + R_2 R_3 + R_2 R_L}$$

neglete R_3

$$I_L = \frac{V R_2}{R_1 R_2 + R_3 R_1 + R_L R_1 + R_2 R_3 + R_2 R_L}$$

case ① To find I_L short R_L



~~applying~~ ~~~~ current divider

$$I_N = \frac{I_T \cdot R_2}{R_2 + R_3} \quad \dots \text{(5)}$$

$$I_N = \frac{V}{R_T} \quad \dots \text{(6)}$$

$$R_T = R_1 + (R_2 \| R_3)$$

$$R_T = R_1 + \frac{(R_2 R_3)}{R_2 + R_3}$$

$$R_T = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2 + R_3} \quad \dots \text{(7)}$$

subs (7) in (6)

$$I_T = \frac{V(R_2 + R_3)}{R_1 R_2 + R_1 R_3 + R_2 R_3} \quad \dots \text{(8)}$$

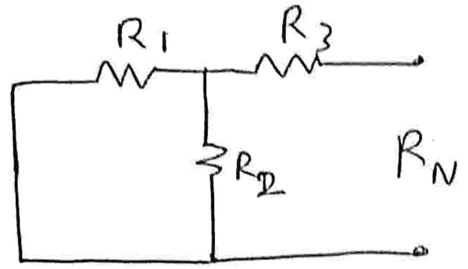
subs (8) in (5)

$$I_N = \frac{\frac{V\cancel{(R_2 + R_3)}}{R_1 R_2 + R_1 R_3 + R_2 R_3} \cdot R_2}{\cancel{R_2 + R_3}}$$

$$\boxed{I_N = \frac{V R_2}{R_1 R_2 + R_1 R_3 + R_2 R_3}} \quad \dots \text{(9)}$$

To find R_N

Voltage source shorted, current open, R_L load removed.



$$R_N = (R_1 || R_2) + R_3$$

$$R_N = \frac{R_1 R_2}{R_1 + R_2} + R_3$$

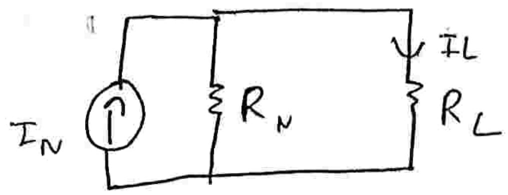
$$\boxed{R_N = \frac{R_1 R_2 + R_3 R_1 + R_2 R_3}{R_1 + R_2}}$$

$$I_L = \frac{I_N \times R_N}{R_N + R_L}$$

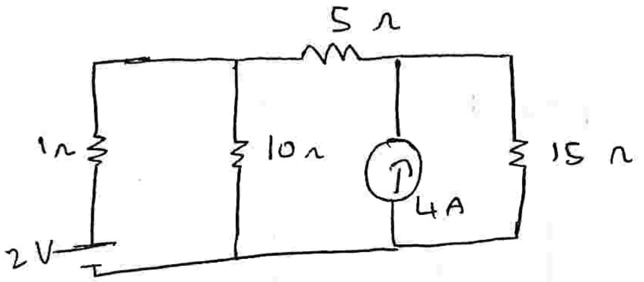
$$I_L = \frac{\dfrac{V R_2}{\cancel{(R_1 R_2 + R_1 R_3 + R_2 R_3)}} \times \dfrac{\cancel{(R_1 R_2 + R_3 R_1 + R_2 R_3)}}{\cancel{(R_1 + R_2)}}}{\dfrac{R_1 R_2 + (R_3 R_1) + R_2 R_3}{\cancel{(R_1 + R_2)}} + R_L (R_1 + R_2)}$$

$$I_L = \frac{V R_2}{R_1 R_2 + R_3 R_1 + R_L R_1 + R_2 R_3 + R_2 R_L}$$

The Norton equivalent circuit is

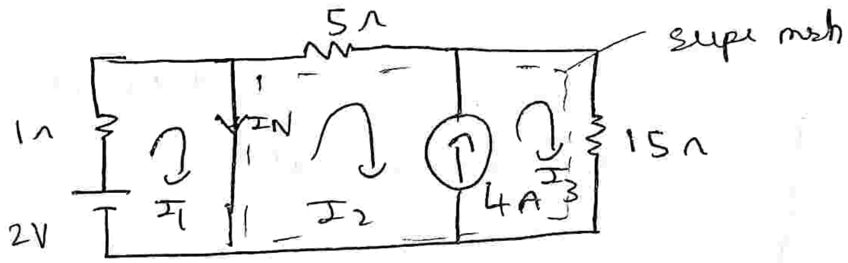


1) using Nortons theorem find the current through 10Ω resistance



soln:
$R_L = 10\Omega$

short 10Ω Replace by I_N



from cRA

$\boxed{I_N = I_1 - I_2}$ ... ①

$I_1 = \frac{V}{R} = \frac{2}{1} = 2\,A$

$\boxed{I_3 - I_2 = 4}$ ... ②

apply KVL to super mesh.

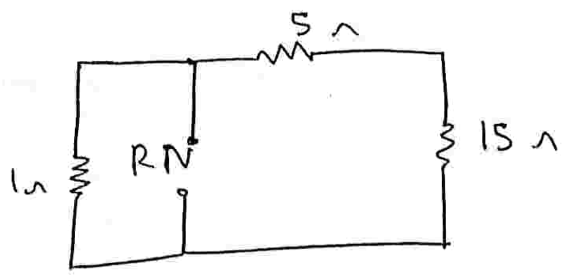
$-5I_2 - 15I_3 = 0$ ... ③

$\boxed{I_2 = -3}$ $\boxed{I_3 = 1}$

$\therefore I_N = 2 - (-3)$

$\boxed{I_N = 5\,A}$

To find RN



$5+15 \parallel 1$

$R_N = \frac{20}{21}$

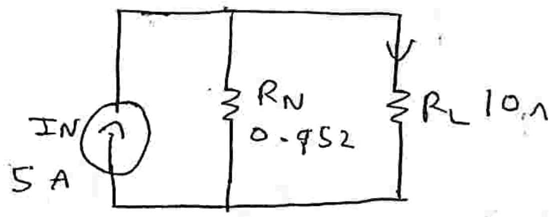
$\boxed{R_N = 0.952\Omega}$

$I_L = \frac{I_N \, R_N}{R_N + R_L}$

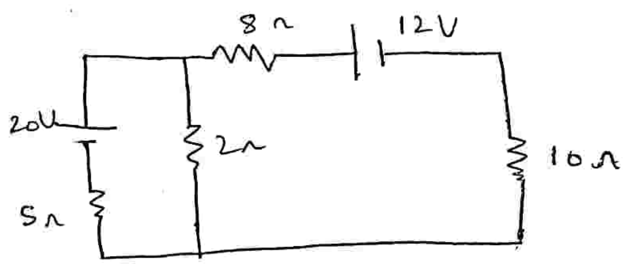
$I_L = \frac{5 \times 0.952}{0.952 + 10}$

$\boxed{I_L = 0.43\text{ A}}$

Norton's equivalent ckt



2)



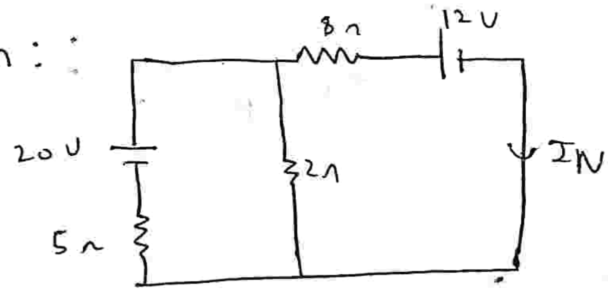
Find the current through 10Ω Resistor through RL

$I_N = -0.666\text{ A}$

$R_N = 9.420\,\Omega$

$I_L = -0.326\text{ A}$

Soln:



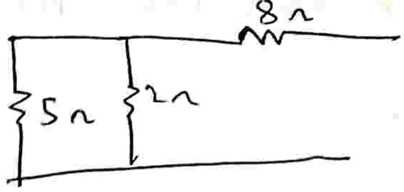
$-5I_1 - 2I_1 + 2I_2 = -20 \quad \ldots (1)$

$-2I_2 + 2I_1 - 8I_2 = 12 \quad \ldots (2)$

$\boxed{I_1 = 2.66}\ \boxed{I_2 = -0.666\text{ A}}$

$\therefore \boxed{I_N = -0.666\text{A}}$

To $R_N =$



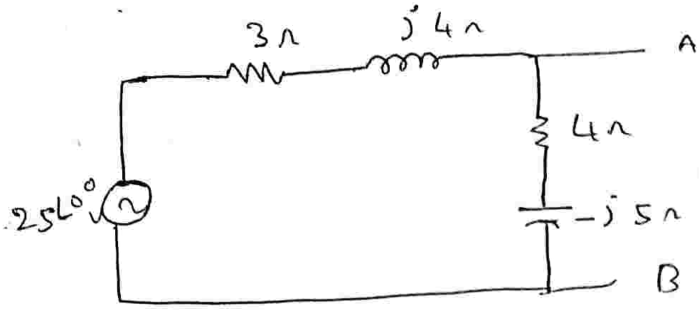
$$R_N = (5 \parallel 2) + 8$$

$$\boxed{R_N = 9.428\,\Omega}$$

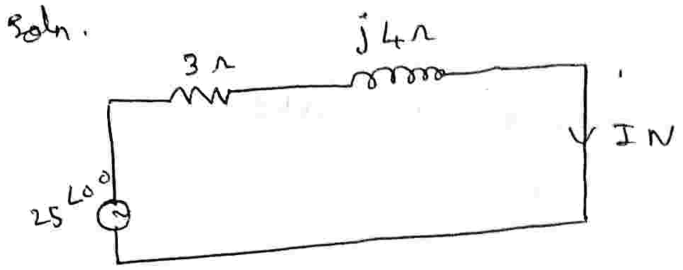
$$I_L = \frac{R_N \times I_N}{R_N + R_L} = \frac{9.428 \times (-0.666)}{9.428 \times 10}$$

$$\boxed{I_L = -0.3231\,A}$$

3) obtain the Norton equivalent for the n/w shown



Soln.



$$-3I_1 - j4I_1 = 25$$

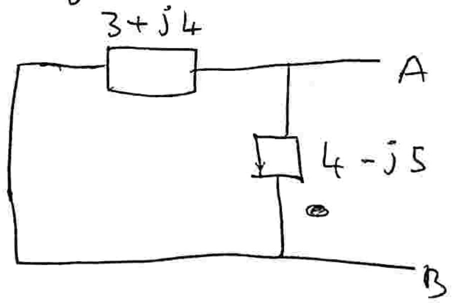
$$(-3-4i)I_1 = 25$$

$$I_1 = \frac{25}{-3-4i}$$

$$\boxed{I_N = -3+4i} \qquad \boxed{I_N = 3-4i}$$

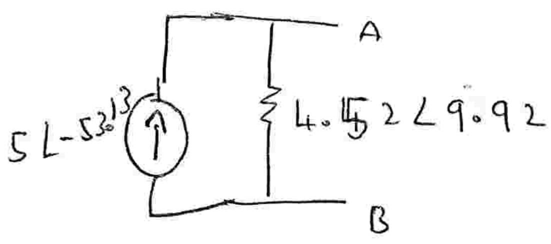
$$\boxed{I_1 = 5\angle -53.13}$$

To find RN

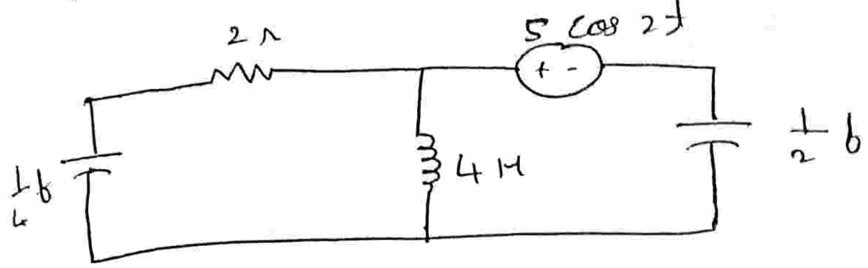


$$R_N = 4.52 \angle 9.92$$

$$\boxed{R_N = 4.46 + 0.78i}$$



4) Determine the current through 4 H for the ckt shown using Nortons theorem.



Represent the given ckt in frequency domain ω = 2 Radians / sec.

$X_L = j\omega L$

$X_L = j \times 2 \times 4$

$\boxed{X_L = j8\,\Omega}$

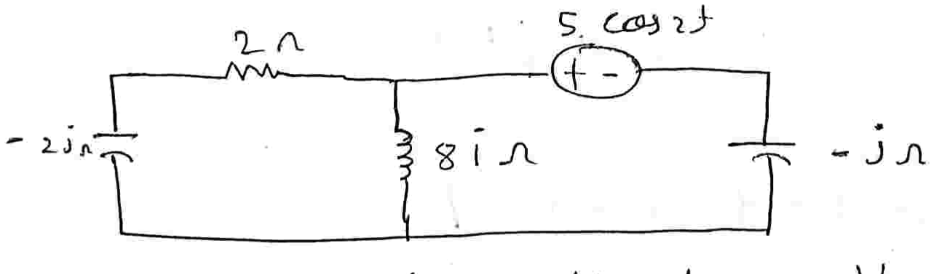
$X_C = \frac{1}{j\omega C}$

$X_C = \frac{1}{j \times 2 \times 0.5}$

$\boxed{X_C = -j\,\Omega}$

$X_C = \frac{1}{j \times 2 \times 0.25}$

$\boxed{X_C = -2j\,\Omega}$

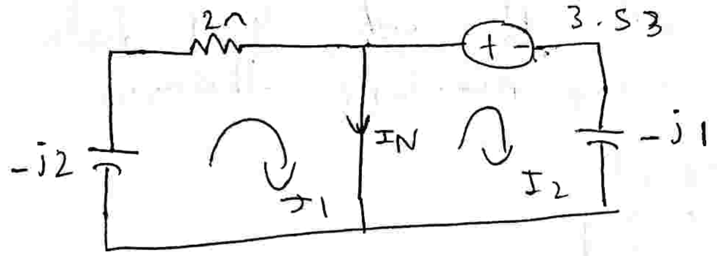


5 cos 2t is of the form $V = V_m \cos \omega t$

$V_m = 5$

$\omega = 2$

$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{5}{\sqrt{2}} = 3.53\ \Omega$



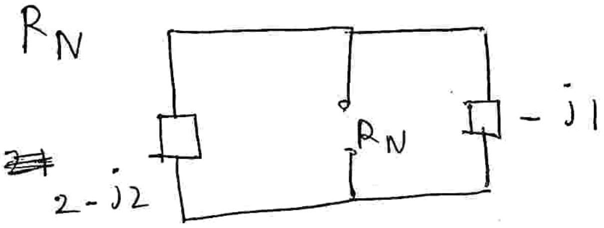
$I_N = I_2 - I_1$

$I_1 = 0$

$I_2 = \frac{V}{R} = \frac{3.53}{-j1}$

$\boxed{I_N = 3.53i}$ or $\boxed{I_N = 3.53\angle 90}$

R_N



$R_N = \frac{(2-j2) \times (-j1)}{(2-j2)+(-j1)}$

$\boxed{R_N = 0.153 - 0.76i}$ $\boxed{R_N = 0.78\angle -78.6}$

$I_L = \frac{R_N \times I_N}{R_N + R_L}$

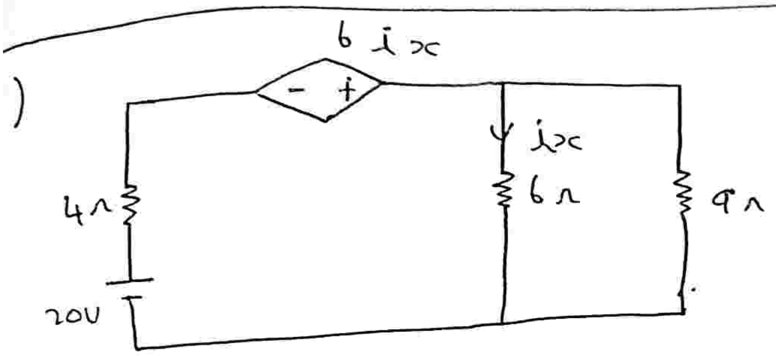
$I_L = \frac{(0.78\angle -78.69)(3.53\angle 90)}{(0.78\angle -78.69)+(8i)}$

$\boxed{I_L = 0.380\angle -77.47^\circ A}$

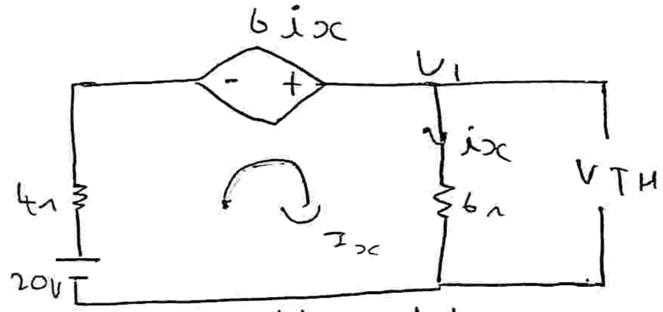
→ Thevenin's & Norton's theorem for the ckt with dependent sources:

- Identify and remove the load resistance
- look into the load terminal & calculate V_{TH}
- place a short on load terminal calculate the I_N & determine $R_{TH} = \frac{V_{TH}}{I_N}$
- The load current $I_L = \frac{V_{TH}}{R_{TH} + R_L}$

)



Determine the current through 9Ω



from the ckt
$V_{TH} = 6\,i_x$

$$20 - 4I_x + \cancel{6I_x} - \cancel{6I_x} = 0$$

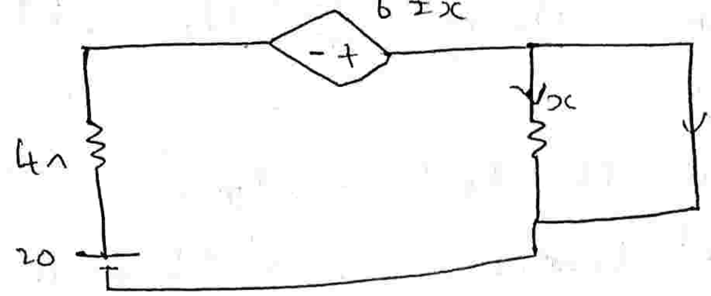
$$20 = 4I_x$$

$$I_x = \frac{20}{4}$$

$$\boxed{I_x = 5A}$$

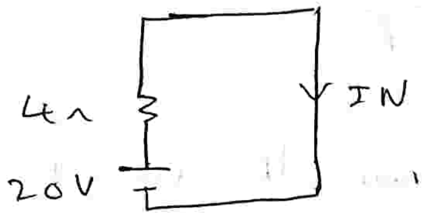
$$V_{TH} = 6 \times 5$$

$$\boxed{V_{TH} = 30V}$$



since R is connected is || with short ckt here

$I_x = 0 \therefore 6 I_x = 0$



$$I_N = \frac{20}{4} = 5$$

$$\boxed{I_N = 5A}$$

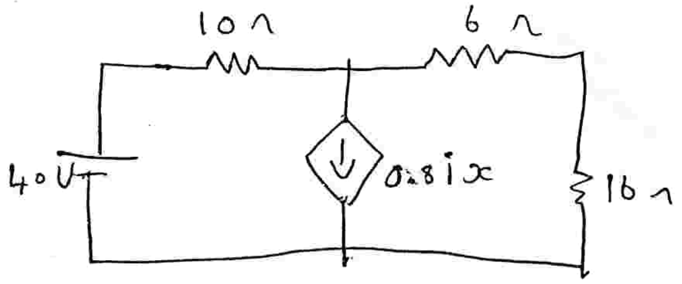
$$R_{TH} = \frac{\cancel{30}^{6}}{\cancel{5}}$$

$$\boxed{R_{TH} = 6\Omega}$$

$$I_L = \frac{30}{6+9}$$

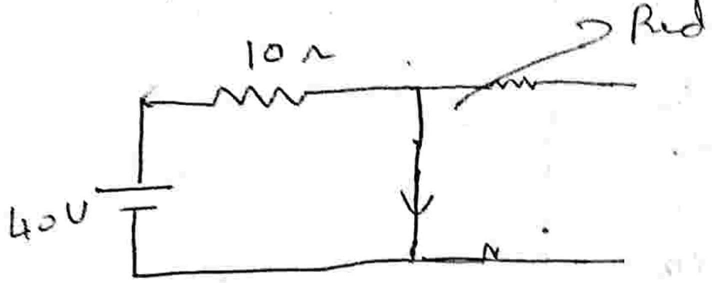
$$\boxed{I_L = 2A}$$

2) Determine the current through 16Ω Resistor using thevenins theorem



To find VTH since $ix = 0$ $0.8\,ix = 0$

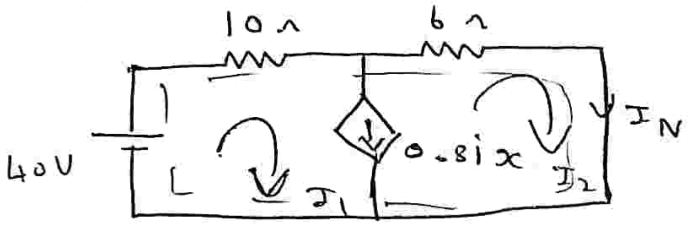
Remove 16 Ω



$$VTH = \frac{40 \times 1\cancel{0}}{1\cancel{0}}$$

$$\boxed{VTH = 40\,V}$$

To find



$\cancel{40 - 10 I_1 - 0.8 I_2 + 0.8 I_1} = 0$

$ix = I_2$

$$\boxed{I_1 - I_2 = 0.8\,ix}$$

$I_1 - 1.8 I_2 = 0$ . . . ①

$40 - 10 I_1 - 6 I_2 = 0$

$$\boxed{40 = 10 I_1 + 6 I_2 = 0}$$ . . . ②

$I_1 = 3$ $\boxed{I_2 = 1.66}$

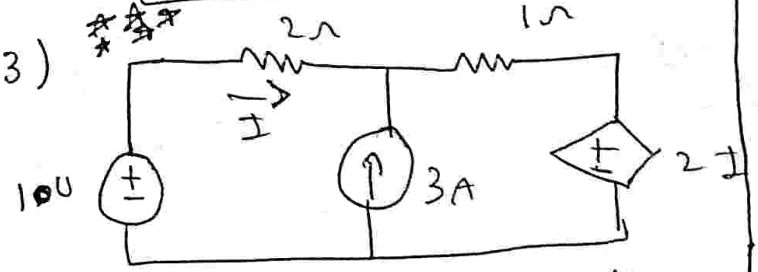
$$\boxed{I_2 = I_N = 1.66\,A}$$

$R_{TH} = \frac{40}{1.66}$ $\boxed{R_{TH} = 24.09\,\Omega}$ $I_L = \frac{V_{TH}}{R_{TH} + R_L}$

$\boxed{I_L = 0.997\,A}$

Thevenin's theorem for both dependent and Independent sources.

3) ***



Find the current I for n/w shown using thevenin's theorem

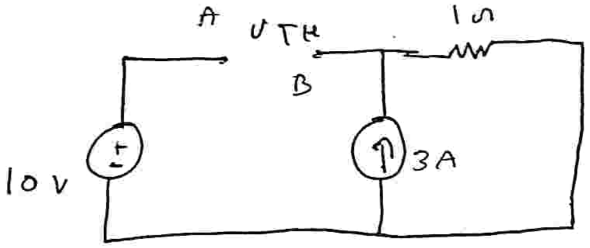
$R_L = 2\Omega$

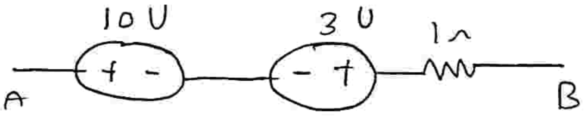
To find V_{th} open load

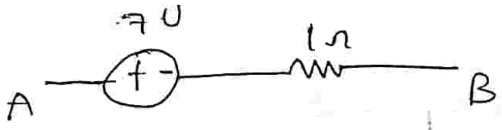
Since 2Ω is removed $I = 0$

$\therefore 2I = 0$

here I it shorted

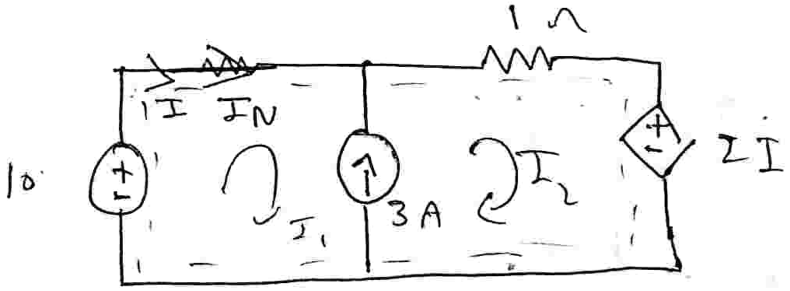






$\therefore V_{TH} = 7V$

to find In short load mark IN



$I = I_1$

$I_2 - I_1 = 3 \quad \ldots (1)$

appl KVL to super mesh

$-1I_2 - 2I_1 = -10$

$$\boxed{I_2 = 5.33A}$$

$$\boxed{I_1 = 2.33A}$$

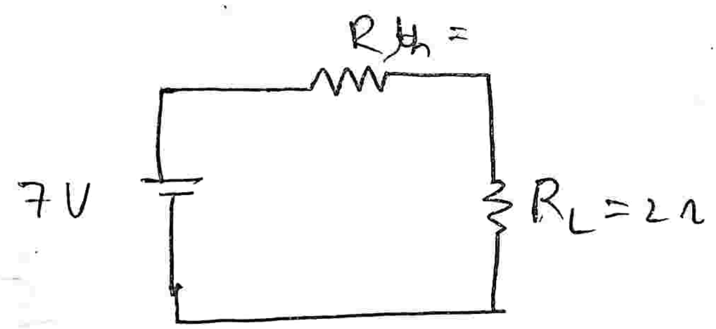
$$\boxed{\therefore IN = 2.33A}$$

$$R_{th} = \frac{V_{Th}}{IN} = \frac{7}{2.33}$$

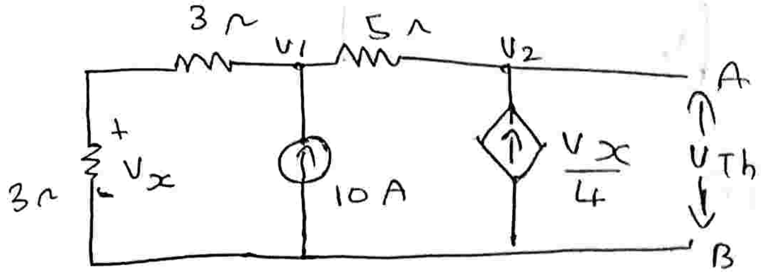
$$\boxed{R_{th} = 3\,\Omega}$$

$$i_L = \frac{V_{th}}{R_{th} + R_L} = \frac{7}{3+2}$$

$$\boxed{i_L = 1.4A}$$



4) Find the thevenin's equivalent for the circuit shown



apply KCL at node.

$V_{th} = V_2$

$$\boxed{V_x = 0.5 V_1}$$

$$\frac{V_1}{6} + \frac{V_1 - V_2}{5} = 10$$

$$V_1 \left(\frac{1}{6} + \frac{1}{5} \right) - \frac{V_2}{5} = 10$$

$$0.366 V_1 - 0.2 V_2 = 10 \quad \ldots (1)$$

$$\frac{V_2 - V_1}{5} - \frac{V_x}{4} = 0$$

$$0.2 V_2 - 0.2 V_1 = \frac{0.5 V_1}{4}$$

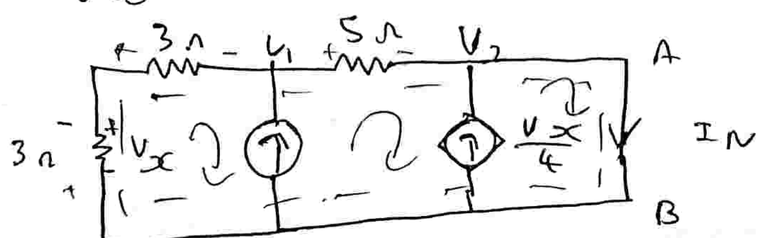
$$-0.2 V_1 - 0.125 V_1 + 0.2 V_2 = 0 \quad \ldots (2)$$

$$-0.325 + 0.2 V_2 = 0 \quad \ldots (3)$$

$$\boxed{V_1 = 240.38} \qquad \boxed{V_2 = 390.625}$$

$$\boxed{V_{Th} = V_2 = 390.62 V}$$

To find IN short terminals A & B



$V_x = -I_1 R$

$\boxed{V_x = -I_1 \cdot 3}$ $\boxed{V_x = -3I_1}$

~~appl~~ ① Super mesh

$I_2 - I_1 = 10 \quad \ldots (1)$

② super mesh.

$$I_3 - I_2 = \frac{V_x}{4}$$

$$I_3 - I_2 = -\frac{3I_1}{4}$$

$$+0.75I_1 - I_2 + I_3 = 0 \quad \ldots (2)$$

apply KVL to super mesh.

$$-6I_1 - 5I_2 = 0 \quad \ldots (3)$$

$\boxed{I_1 = -4.54A}$ $\boxed{I_2 = 5.45\ A}$ $\boxed{I_3 = 8.86A}$

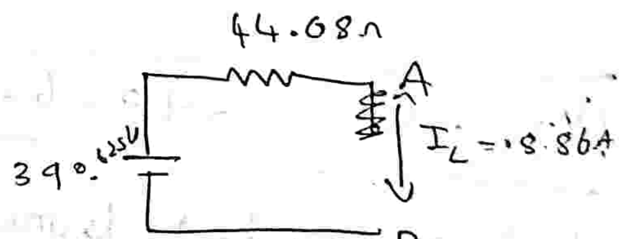
$$\boxed{\therefore I_N = 8.86A}$$

$$R_{th} = \frac{V_{th}}{I_N} = \frac{390.625}{8.86}$$

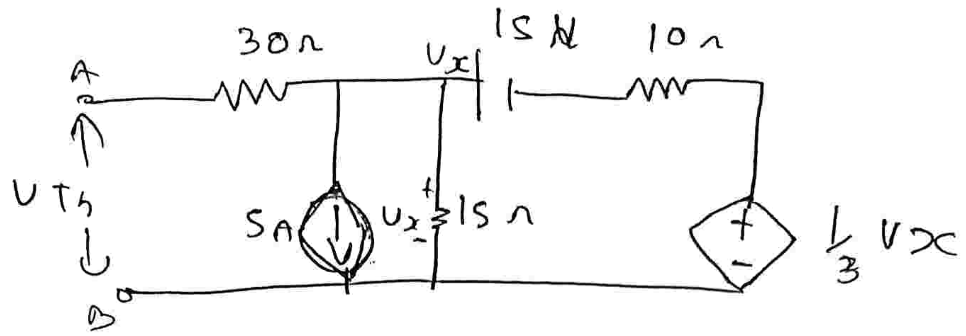
$$\boxed{R_{th} = 44.08\Omega}$$

$$I_L = \frac{V_{th}}{R_{th}}$$

$$\boxed{I_L = 8.86A}$$

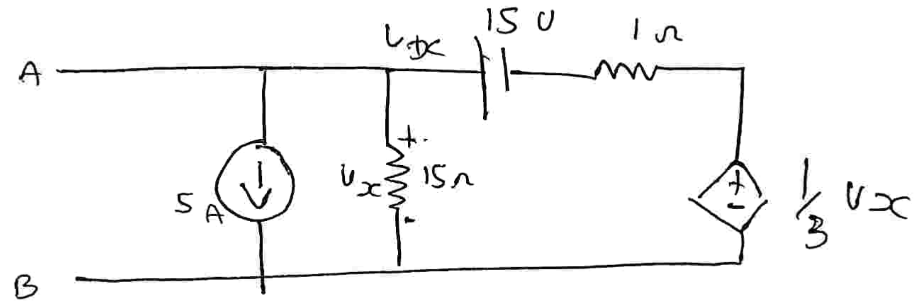


5) obtain the thevenin equivalent for the ckt shown



To find VTH.

since the load terminals is open current I =0 hence 30Ω is redundant



apply KCL

$$\frac{V_x}{15} + \frac{V_x - 15 - \frac{1}{3}V_x}{10} = -5$$

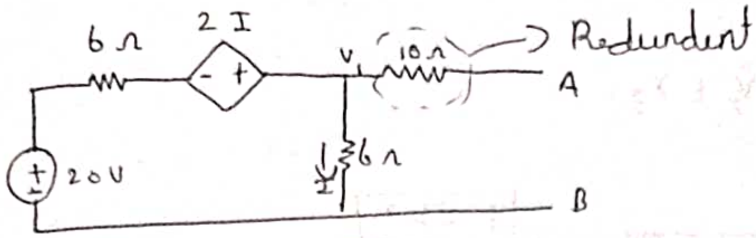
$$V_x\left(\frac{1}{15} + \frac{1}{10} + \frac{\frac{1}{3}(1)}{10}\right) = 1.5 - 5$$

$$V_x(0.2) = 1.5 - 5$$

$$V_x = \frac{1.5 - 5}{0.2}$$

$$\boxed{V_x = -17.5V}$$

1)



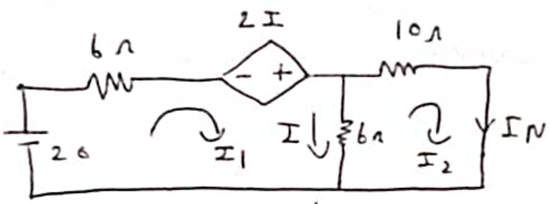
$V_{Th} = V_1$

$20 - 6I_1 + 2I_1 - 6I = 0$

$20 - 10I = 0$

$I = \frac{20}{10}$ $\boxed{I = 2A}$

$V_{Th} = I \times R = 2 \times 6 = 12V$



apply KVL to loop 1

$I = I_1 - I_2$

$-10I_2 - 6(I_2 - I_1) = 0$

$-10I_2 - 6I_2 + 6I_1 = 0$

$6I_1 - 16I_2 = 0$

apply KVL to loop ②

$20 - 6I_1 + 2I - 6(I_1 - I_2) = 0$

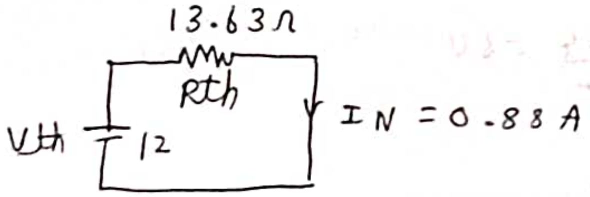
$20 - 10I_1 + 4I_2 = 0$

$10I_1 - 4I_2 = 20$

$I_1 = 2.35 A$ $I_2 = 0.88 A$

$I_N = I_2 = 0.88A$ //

$R_{th} = \frac{12}{0.88} = 13.63\Omega$



2) $V_1 = 20, \quad V_2 = 40, \quad V_3 = 50, \quad Z_1 = 2, \quad Z_2 = 4, \quad Z_3 = 5$

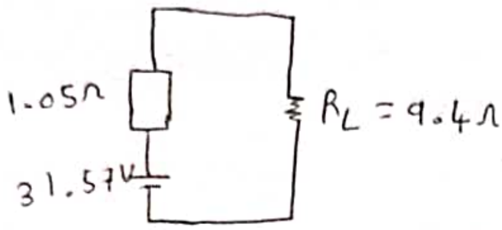
$Y_1 = 0.5, \quad Y_2 = 0.25, \quad Y_3 = \frac{1}{5} = 0.2$

$$Z_m = \frac{1}{Y_1 + Y_2 + Y_3}$$

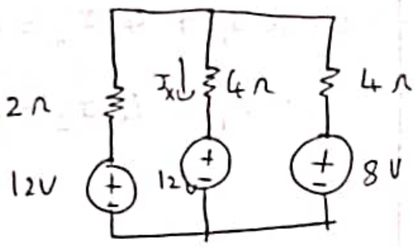
$$Z_m = \frac{1}{0.5 + 0.25 + 0.2} = \boxed{1.05\Omega}$$

$$V_m = \frac{20(0.5) + 40(0.25) + 50(0.2)}{0.5 + 0.25 + 0.2}$$

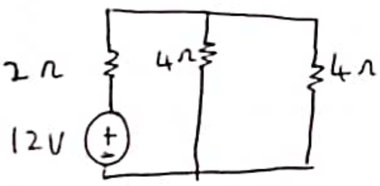
$$\boxed{V_m = 31.57V}$$



3) Find I_x using Super position theorem:



i) consider 12V alone

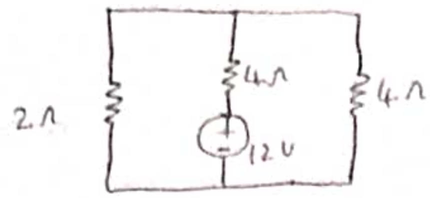


4||4

$$V_1 = \frac{2(12)}{2+2} = \frac{24}{4} = 6V$$

$$I_x' = \frac{V}{R} = \frac{6}{4} = 1.5A$$

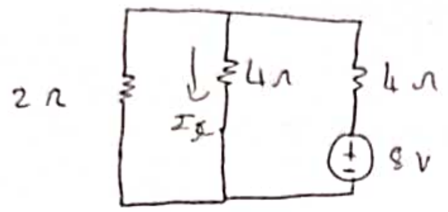
ii) consider 12 V alone



$4 \| 2 = 1.33\Omega$

$$I = \frac{V}{R} = -\frac{12}{4+1.33}$$ $\boxed{I = -2.25A}$

iii) consider 8 V alone



$$V = \frac{8 \times V_1}{4+2}$$

$4 \| 2 = 1.33\Omega$

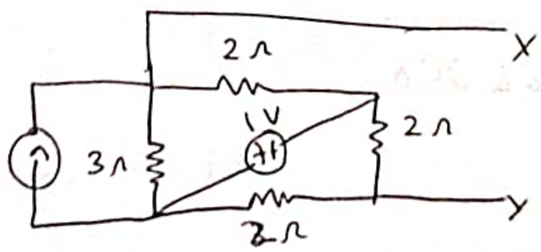
$$VDR = \frac{8(1.33)}{5.33} = 2V$$

$$\boxed{I = \frac{V}{R} = 0.5 A}$$

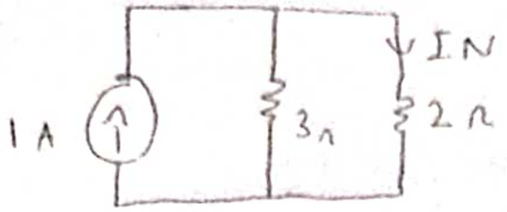
$I_T = 1.5 - 2.25 + 0.5$

$\boxed{I_T = -0.25A}$

+) Find the current through $R_L = 1\Omega$

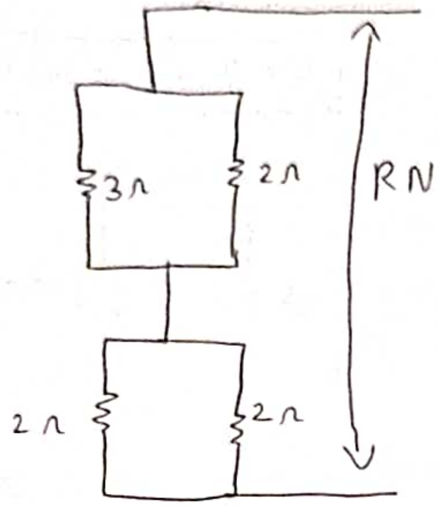


To find I_N, short R_L & replace with I_N



$$I_N = \frac{3(1)}{2+3} = 0.6A$$

To find R_N, open load



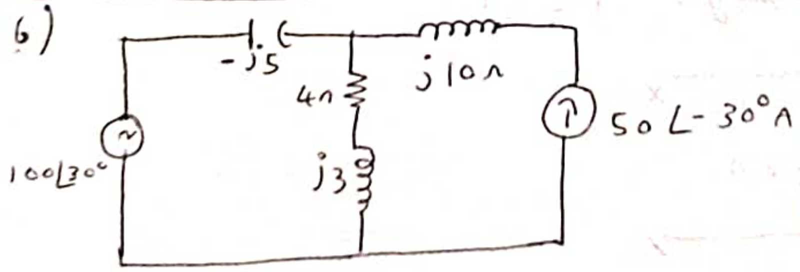
$$R_N = (3 \| 2) + (2 \| 2) = 2.2\Omega$$

for $R_L = 1\Omega$

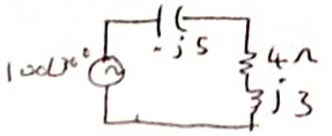
$$I_L = \frac{I_N R_N}{R_N + R_L} = \frac{(0.6 \times 2.2)}{2.2+1} = 0.41A$$

$$\boxed{I_L = 0.41A}$$

6)

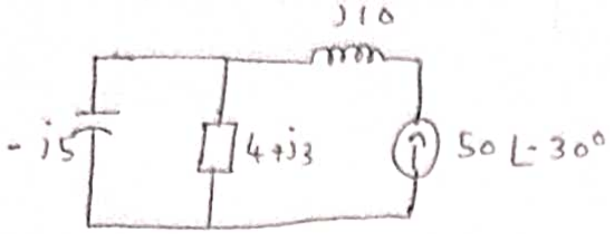


consider only $100\angle 30°$ alone



$$I' = \frac{V}{Z_T} = \frac{15.42 - 98.8i}{(-j5)+(4+j3)} \Rightarrow 12.96 - 18.2i\ A$$

case ii)

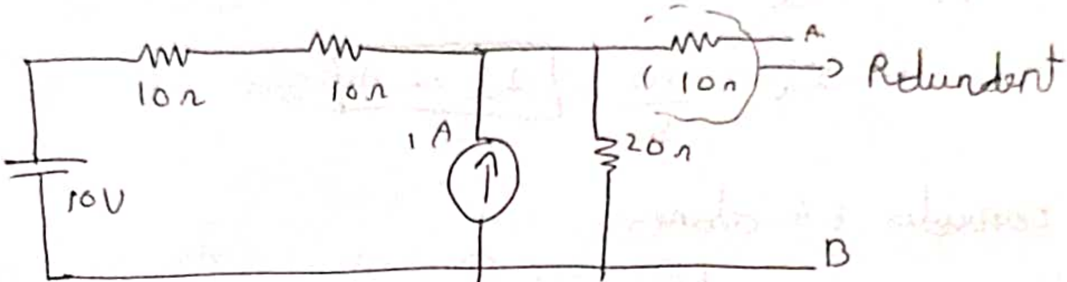


$$I_{(4+j3)} = \frac{-j5(50\angle -30^\circ)}{(-j5)+(4+j3)} = 53.25 + 16.9i$$

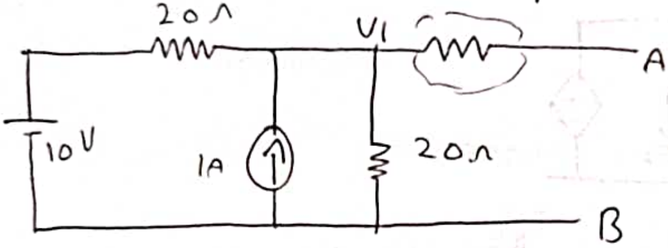
$$\boxed{I_T = I' + I'' = 66.21 - 1.3i}$$

---

7)



⇒



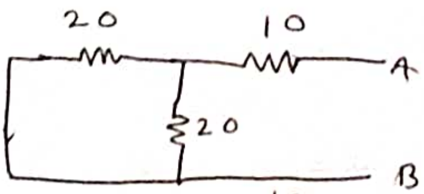
To find V_1 apply KCL

$$\frac{V_1 - 10}{20} - 1 + \frac{V_1}{20} = 0$$

$$0.1V_1 - 1.5 = 0$$

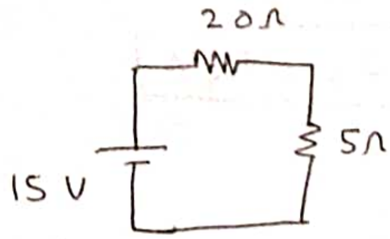
$$V_1 = \frac{1.5}{0.1} \quad \boxed{V_1 = 15V}$$

To find Rth



$10 + 10$

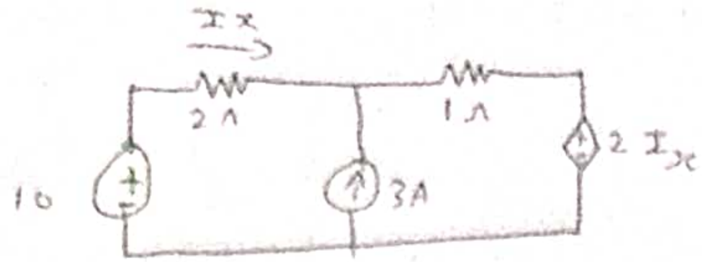
$= 20\,\Omega$



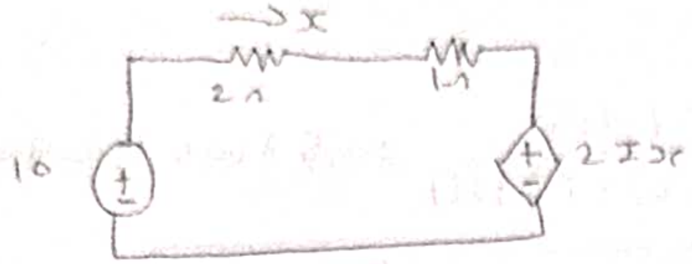
$$V \text{ across } 5\Omega = \frac{5\times 15}{25} = 3V$$

$$\boxed{P_{5\Omega} = \frac{3^2}{5} = 1.8W}$$

8) find I_x using superposition theorem



consider 10V alone

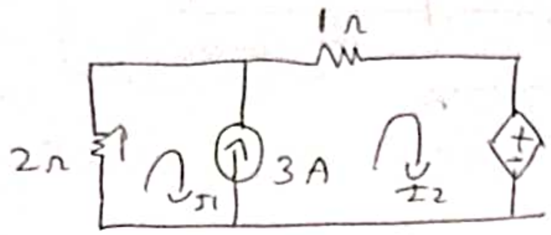


apply KVL

$10 - 2I_x - 1I_x - 2I_x = 0$

$I_x = \frac{10}{5}$ $\boxed{I_x = 2A}$

consider 3A alone



$I_2 - I_1 = 3 \quad \ldots \quad (1)$

$-2I_x - 1I_2 - 2I_x = 0$

$-4I_1 - 1I_2 = 0 \quad \ldots \quad (2)$

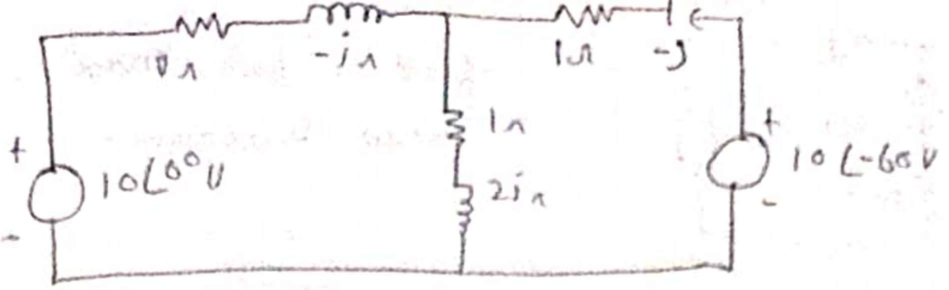
solving 1 & 2

$\boxed{I_1'' = -0.6A}$

$I_x = 2 + (-0.6)$

$\boxed{I_x = 1.4A}$

9)



i) consider 10V

$R_T = (1+2j) \parallel (1-j) + (1-j)$

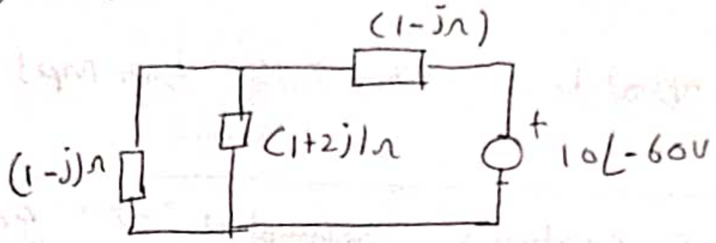
$R_T = (2.4 - 1.2i)\,\Omega$

$I_T = \dfrac{10}{2.4 - 1.2i}$

$I_{Z3} = \dfrac{I_T \cdot Z_2}{Z_2 + Z_3}$

$\boxed{I_T = 3.33 + 1.66i}$

ii) consider $10\angle -60 \Rightarrow -9.52 + 3.04i\,V$



$R_T = 2.4 - 1.2i\,\Omega$

$I_T = \dfrac{V}{R} = \dfrac{-9.52 + 3.04i}{2.4 - 1.2i}$

$I_T = -3.68 - 0.573i$

$\boxed{I_T = (3.72 \angle -2.98)A}$

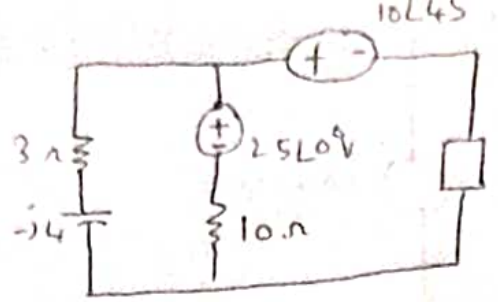
$I_{Z3} = \dfrac{(1-j)(3.72 \angle -2.98)}{(1-j)\,(1+2j)}$

$\boxed{I_{Z3} = 2.35 \angle 2.04}$

$I = I_{Z_3}' + I_{Z_3}''$

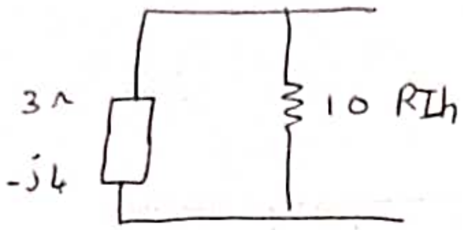
$\boxed{I = 0.73 \angle 0.62°A}$

10)



find Z_L for max Power transmission.

Z_L should be equal to Rth to find Rth, open Z_L, short V source



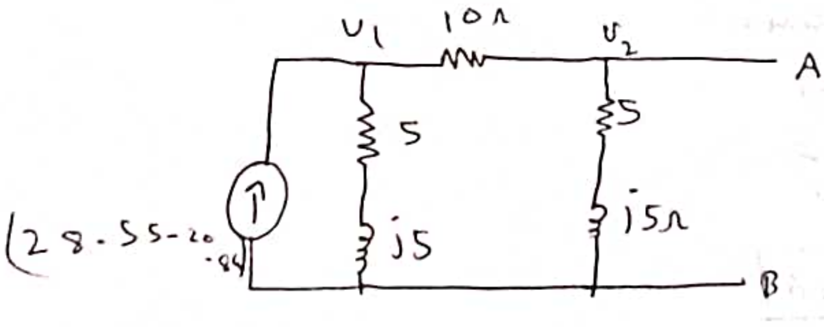
$R_{th} = (3-j4) \parallel 10$

$\boxed{R_{th} = 2.97 - 2.161\,\Omega}$

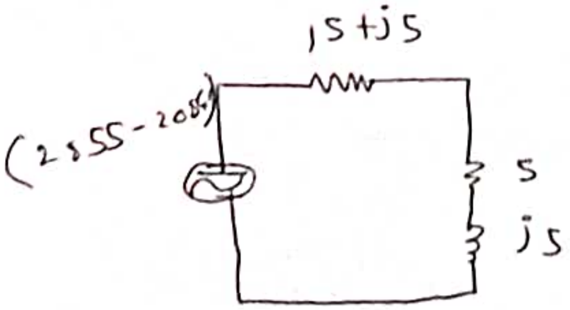
R th should be equal to 2.97 - 2.16i for MPT

---

11) obtain Thevenin's & norton's equivalent ckt find the current through 10Ω RL across A-B



convert to voltage source



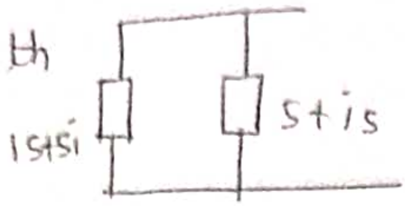
$$V_{th} = \frac{(28.55 - 20.84i)(5+j5)}{(15+5i) + (5+j5)}$$

$V_{th} = 11.17 \angle -0.30$

or

$\boxed{V_{th} = 10.64 - 3.39i}$

R th

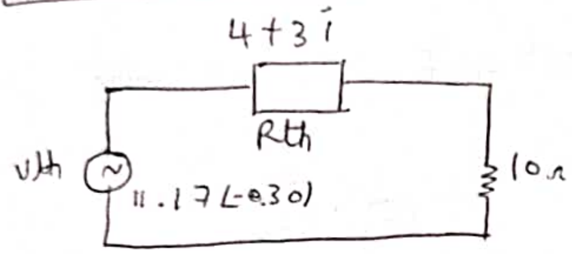


R_{th} $(15+5i) \parallel (5+5i)$

$R_{th} = (4+3i)\Omega$

or

$\boxed{R_{th} = 5 \angle 0.64 \Omega}$



→ Nortons equivalent ckt.

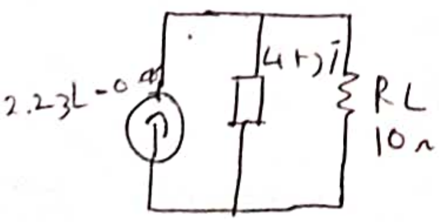
$$R_{th} = \frac{V_{th}}{I_N}$$

$$I_N = \frac{V_{th}}{R_{th}}$$

$$I_N = \frac{10.64 - 3.39i}{5 \angle 0.64}$$

$\boxed{I_N = 2.120 - 0.70i}$

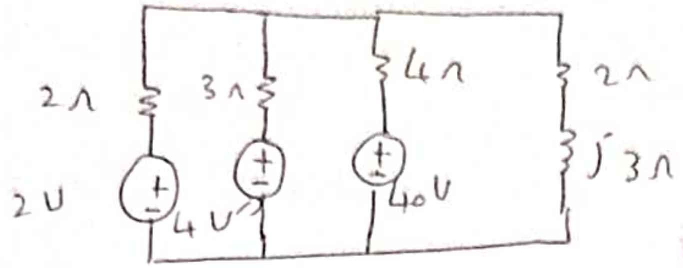
$R_N = R_{th}$



$$I_L = \frac{I_N \cdot R_N}{R_N + R_L} = \frac{(1.29 - 1.8i)(4+3i)}{(4+3i)(10)}$$

$$\boxed{I_L = 0.675 - 0.38i\ A}$$

12) <u>millman's theorem</u>



$Z_1 = 2\Omega \qquad Z_2 = 3\Omega \qquad Z_3 = 4\Omega$

$y_1 = \frac{1}{Z_1} = 0.5\mho \quad y_2 = \frac{1}{Z_2} = 0.33\mho \qquad y_3 = \frac{1}{Z_3} = \frac{1}{4} = 0.25\mho$

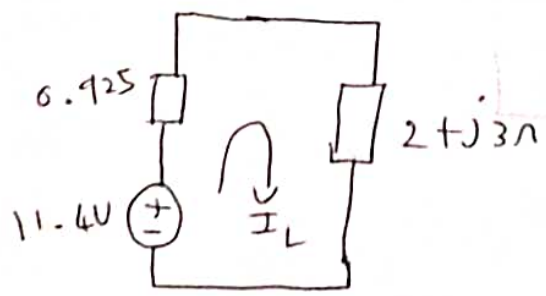
$$Z_m = \frac{1}{0.5 + 0.33 + 0.25}$$

$$\boxed{Z_m = 0.925\Omega}$$

$$V_m = \frac{V_1 y_1 + V_2 y_2 + V_3 y_3}{y_1 + y_2 + y_3}$$

$$V_m = \frac{(2)(0.5) + 4(0.33) + (40)(0.25)}{0.5 + 0.33 + 0.25}$$

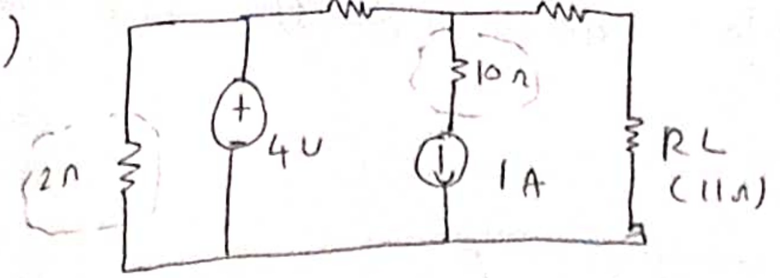
$$\boxed{V_m = 11.4V}$$



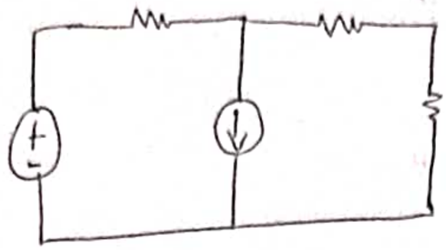
$$I_L = \frac{V_m}{Z_m + 2 + j3} = \frac{11.4}{0.925 + 2 + j3}$$

$$\boxed{I_L = 1.899 - 1.9481\ A}$$

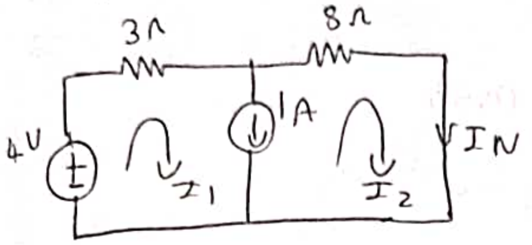
13)



2Ω & 10Ω are redundant from the ckt



To find I_{N_1}



supermesh eqn.

$1 = I_1 - I_2 \quad \ldots (1)$

apply KVL to supermesh

$4 - 3I_1 - 8I_2 = 0$

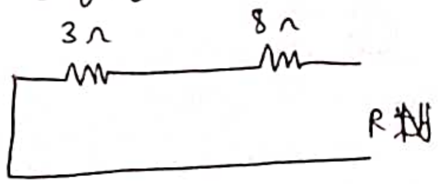
$3I_1 + 8I_2 = 4 \quad \ldots (2)$

from the ckt $I_2 = I_N$

from (1) & (2)

$\boxed{I_1 = 1.09A}$ $\boxed{I_2 = 0.09A}$

To find RN



$R_N = 3+8$

$\boxed{R_N = 11\Omega}$

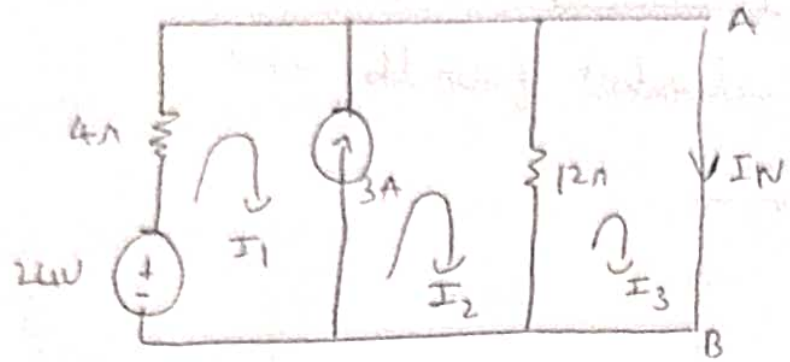
$$I_L = \frac{I_N \cdot R_N}{R_N + R_L}$$

$$I_L = \frac{(0.09)(11)}{11+11}$$

$\boxed{I_L = 0.045A}$

23) Find the Norton's equivalent for the ckt shown

12)



from the ckt $I_3 = I_N$

$$I_2 - I_1 = 3 \quad \ldots (1)$$

applying KVL to super mesh

4

$$24 - 4I_1 - 12I_2 + 12I_3 = 0$$

$$-4I_1 - 12I_2 + 12I_3 = -24 \quad \ldots (2)$$

applying KVL to loop (3)

$$-12I_3 + 12I_2 = 0$$

$$12I_2 - 12I_3 = 0 \quad \ldots (3)$$

solving (1), (2), (3)

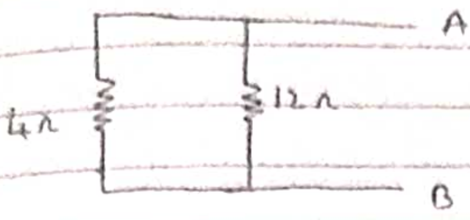
$I_1 = 6A$

$I_2 = 9A$

$I_3 = 9A$

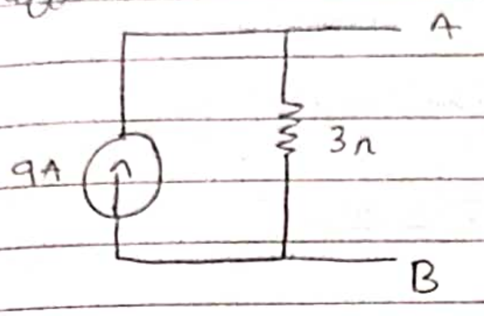
$$\boxed{\therefore I_3 = I_N = 9A}$$

To find RN

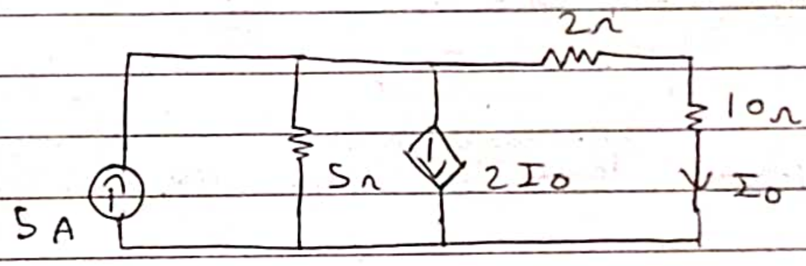


$R_N = 3\,\Omega$

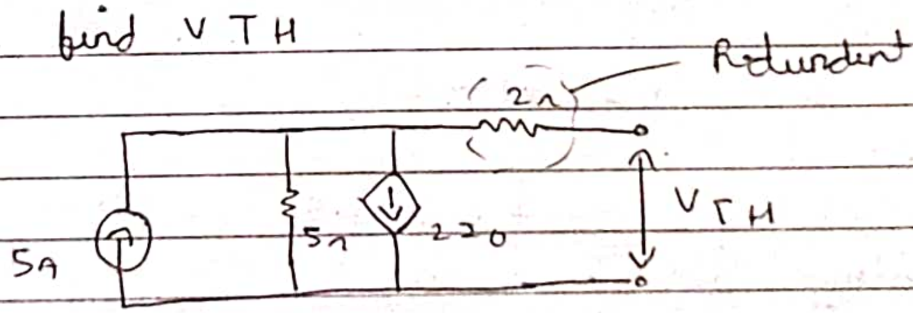
equivalent circuit of norton's



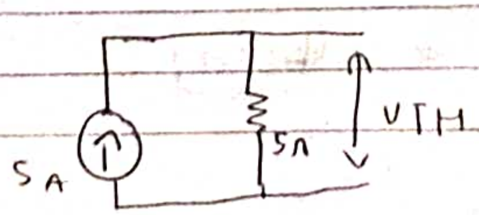
25. In the circuit shown, find the current through $10\,\Omega$ resistor using thevenin's theorem



-> To find V_{TH}



here R_L is open so $I_0 = 0$ hence $2I_0 = 0$



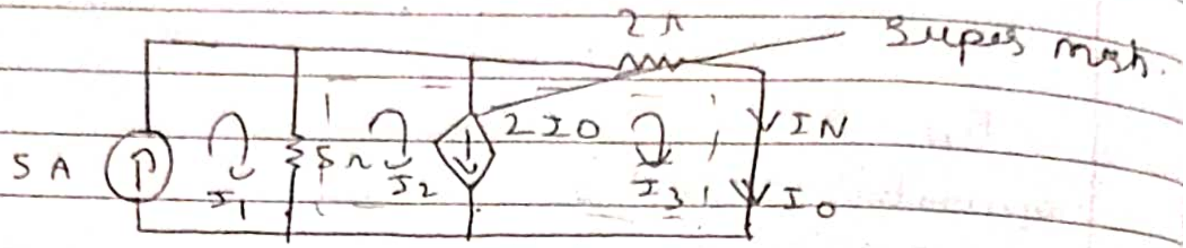
from ohm's law

$V = IR = 5 \times 5$

$\boxed{V = 25\,V}$

$$\boxed{V_{TH} = 25V}$$

To find IN



super mesh equation

$\therefore I_0 =$

$$I_2 - I_3 = 2I_0$$

$$I_2 - I_3 = 2I_3$$

$$I_2 - 3I_3 = 0 \quad \ldots (1)$$

from the ckt

$$\boxed{I_1 = 5A}$$

applying KVL to super mesh.

$$-5I_2 + 5I_1 - 2I_3 = 0$$

$$-5I_2 + 5(5) - 2I_3 = 0$$

$$-5I_2 - 2I_3 = -25 \quad \ldots (2)$$

solving (1) & (2)

$\boxed{I_2 = 4.749 A}$ $\boxed{I_3 = 1.473}$

$$\boxed{\therefore I_N = I_3 = 1.473 A}$$

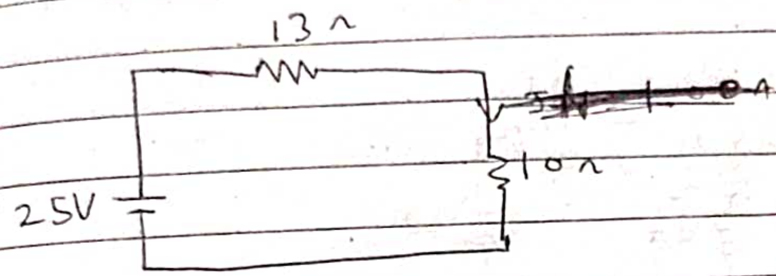
$$\boxed{I_N = 1.473 A}$$

$$R_{TH} = \frac{25}{1.\cancel{47}3}$$

$$\boxed{R_{TH} = 17\,\Omega}$$

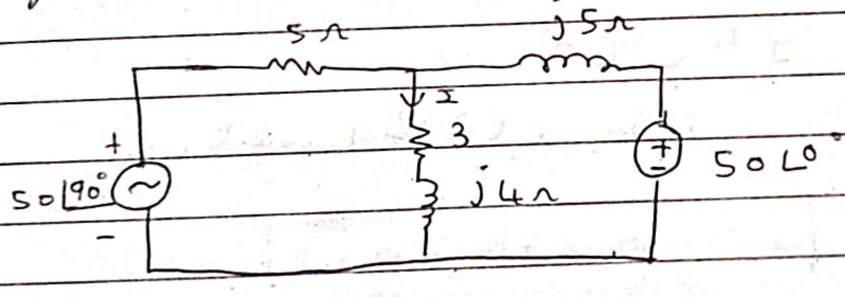
$$I_L = \frac{25}{17+10}$$

$$I_L = 0.92 \approx 1\,A$$

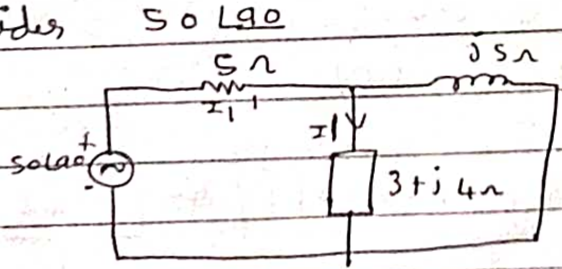


$I_L = 0.92 \approx 1\,A$

26. Find the current through $(3+j4)\,\Omega$ using super position theorem



consider 50∠90



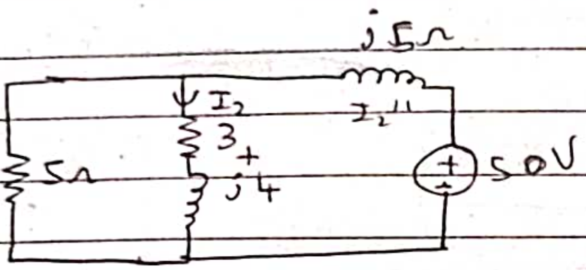
$$I_1 = \frac{I_1' \times j5\,\Omega}{(3+j4)+(j5)}$$

$$I_1' = \frac{50\angle 90^\circ}{5 + (0.833 + 2.5i)}$$

$$\boxed{I_1' = 3.10 + 7.241\,i}$$

$$I_1 = \frac{(3.10 + 7.24i) \times (j5\Omega)}{(3 + j4) + (j(5)}$$

$$\boxed{I_1 = 1.32 + 0.098\,i} \quad \text{or} \quad \boxed{I_1 = 1.33 \angle 4.22}$$



$$I_2 = \frac{5 \times I_2''}{(3 + 4i) + (5)}$$

$$I_2'' = \frac{j50}{(j5\Omega) + (2.5 + 1.25i}$$

$$\boxed{I_2'' = 0.275 - 0.68}$$

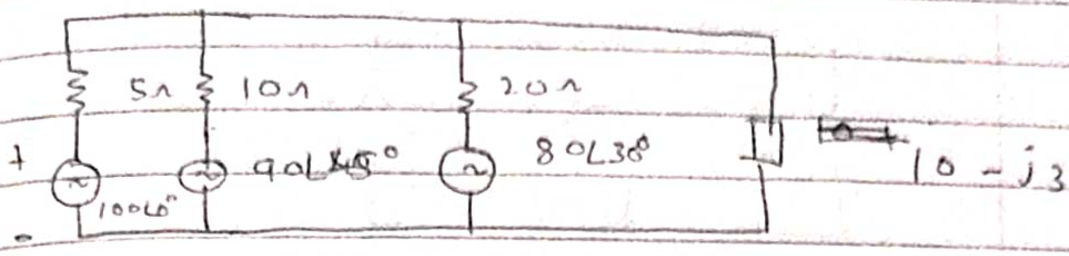
$$I_2 = \frac{5 \times (0.275 - 0.68)}{(3 + 4i) + (5)}$$

$$\boxed{I_2 = -0.202 + 0.1011\,i} \quad \text{or} \quad \boxed{0.22 \angle 153.43\,A}$$

$$I = I_1 + I_2$$

$$\therefore \boxed{I_1 = 1.14 \angle 9.85\,A} \quad \text{or} \quad \boxed{1.12 + 0.196\,i\,A}$$

27) for the ckt shown find the current flowing in $10-j3$ using millman's theorem



$Z_1 = 5 \qquad Z_2 = 10 \qquad Z_3 = 20$

$Y_1 = 0.2 \qquad Y_2 = 0.1 \qquad Y_3 = 0.05$

$$Z_m = \frac{1}{Y_1 + Y_2 + Y_3} = \frac{1}{0.2 + 0.1 + 0.05} = 2.85$$

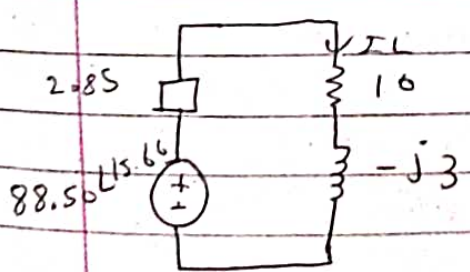
$\boxed{Z_m = 2.85\Omega}$

$\boxed{V_1 = 100} \qquad \boxed{V_2 = 63.63 + 63.63i} \qquad \boxed{V_3 = 69.28 + 40i}$

$$V_m = \frac{V_1 Y_1 + V_2 Y_2 + V_3 Y_3}{Y_1 + Y_2 + Y_3}$$

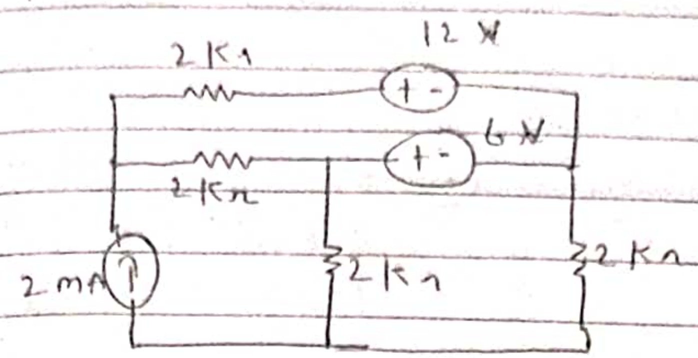
$$V_m = \frac{100(0.2) + 63.63 + 63.63i(0.1) + 69.28 + 40i(0.05)}{0.2 + 0.1 + 0.05}$$

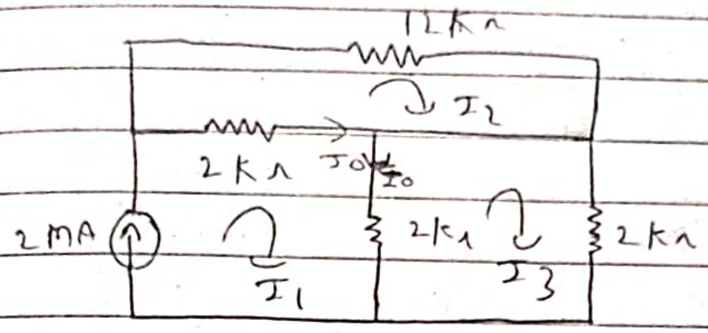
$\boxed{V_m = 85.22 + 23.89i}$ or $\boxed{V_m = 88.50\angle 15.66\,V}$



$\boxed{I_L = 6.70\angle 28.80\,A}$ or $\boxed{I_L = 5.877 + 3.23i\,A}$

29. using superposition theorem to find io in the ckt





$$-4I_1 + 2I_3 = 2 \quad \ldots (1)$$

$$-12I_2 - 12I_2 + 12I_1 = 0 \quad \ldots (2)$$

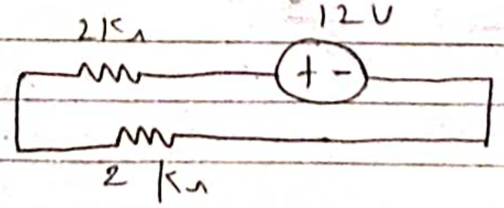
$$-2I_3 - 2I_1 - 2I_3 = 0 \quad \ldots (3)$$

$\boxed{I_1 = -0.4\text{mA}}$ $\boxed{I_2 = -0.2\text{mA}}$ $\boxed{I_3 = 0.2\text{mA}}$

$$I_1 = I_1' = -0.4\text{ mA}$$

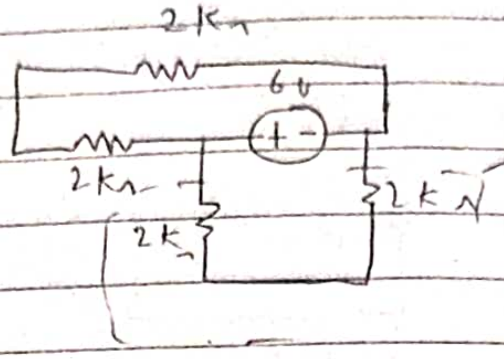
Case (2)

12V acting alone.



$$I = \frac{V}{R} = \frac{12}{4K}$$ $\boxed{I_1'' = 3\text{ mA}}$

(Voltage alone:



Redundant as they are ||le with

$$I = \frac{V}{R} = \frac{6}{4} = 1.5\ mA$$

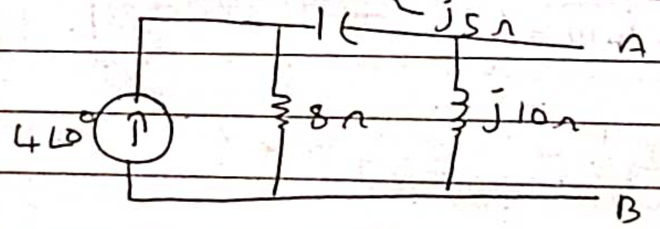
$$\boxed{I_1''' = 1.5\ mA}$$

$$I_0 = I_1' + I_1'' + I_1'''$$

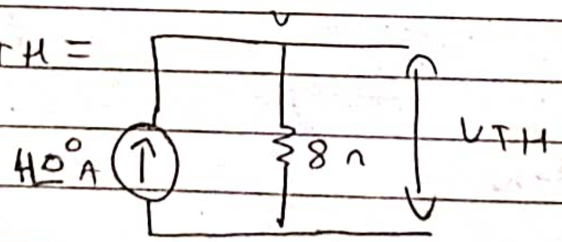
$$I_0 = -0.4mA + 3mA + 1.5mA$$

$$\boxed{I_0 = 4.1\ mA}$$

30) Find the thevenin & norton's equivalent ckt a

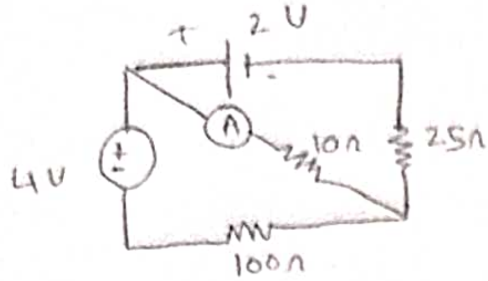


$V_{TH} =$



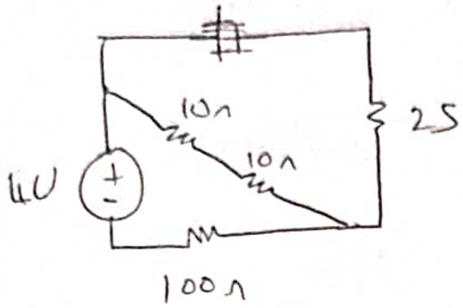
$$\boxed{V_{TH} = 32V}$$

21) Determine I through an Ammeter having internal resistance



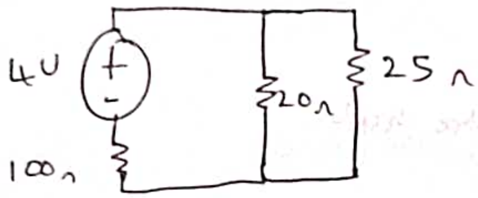
→ consider current through ammeter as, I

cas 1: 4V acting alone.



$R = 10 + 10$

$= 20\,\Omega$

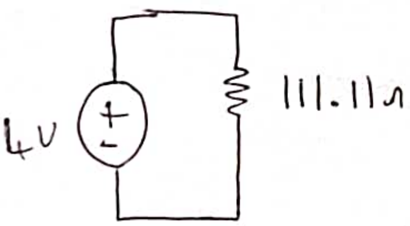


$R = 20 \parallel 25$

$R = 11.11\,\Omega$

$R = 11.11 + 100$

$R = 111.11$



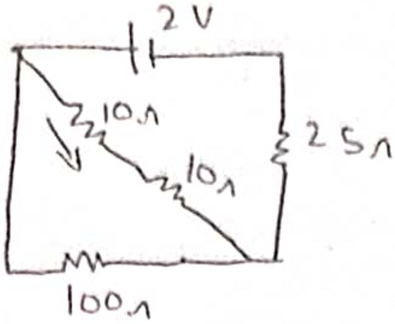
$$I = \frac{V}{R} = \frac{4}{111.11} = 0.036\,A$$

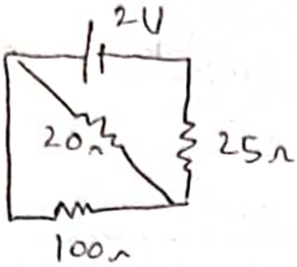
Apply current divider rule.

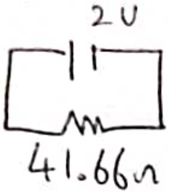
$$I_1 = \frac{I \times 25}{20 + 25}$$

$$\boxed{I_1 = 20\,mA}$$

case 2: 2V acting alone.







$$I = \frac{2}{41.66}$$

$$\boxed{I = 48\,mA}$$

applying current divider rule.

$$I_2 = \frac{I \times 100}{20 + 100}$$

$$\boxed{I_2 = 40\,mA}$$

Current through ammeter $= I_1 + I_2$

$= 20m + 40m$

$$\boxed{I = 60\,mA}$$